

梦想掌控：通过潜在想象学习行为

Danijar Hafner* Timothy Lillicrap Jimmy Ba Mohammad Norouzi 多伦多大学
DeepMind 多伦多大学 Google Brain Google Brain

摘要

学习的世界模型总结智能体的经验，以促进复杂行为的学习。虽然通过深度学习从高维感官输入中学习世界模型正变得可行，但从中衍生出行为的方式有很多种可能。我们提出了Dreamer，一个通过纯潜在想象从图像中解决长时域任务的强化学习智能体。我们通过将学习的状态价值的解析梯度反向传播回在紧凑状态空间中想象的轨迹，来高效地学习行为。在20个具有挑战性的视觉控制任务中，Dreamer在数据效率、计算时间和最终性能方面均超越了现有方法。

1 引言

智能体能够在复杂环境中实现目标，即使它们从未两次遇到完全相同的情况。这种能力需要从过往经验中构建世界的表示，从而能够泛化到新情境。世界模型提供了一种显式方式，通过一个能够预测未来的参数化模型来表示智能体对世界的知识。

当感官输入为高维图像时，潜在动力学模型可以抽象观测结果，在紧凑状态空间中进行前向预测（Watter等人, 2015; Oh等人, 2017; Gregor等人, 2019）。与图像空间中的预测相比，潜在状态具有较小的内存占用，支持并行想象数千条轨迹。通过深度学习和潜变量模型的进步（Krishnan等人, 2015; Karl等人, 2016; Doerr等人, 2018; Buesing等人, 2018），学习有效的潜在动力学模型正变得可行。行为可以通过多种方式从动力学模型中导出。通常，想象奖励会通过参数化策略（Sutton, 1991; Ha和Schmidhuber, 2018; Zhang等人, 2019）或在线规划（Chua等人, 2018; Hafner等人, 2018）来最大化。然而，仅考虑固定想象范围内的奖励会导致短视行为（Wang等人, 2019）。此外，先前的工作通常采用无导数优化来增强对模型误差的鲁棒性（Ebert等人, 2017; Chua等人, 2018; Parmas等人, 2019），而非利用神经网络动力学提供的解析梯度（Henaff等人, 2019; Srinivas等人, 2018）。

我们提出 Dreamer 智能体，该智能体仅通过潜在想象从图像中学习长时域行为。一种新颖的行者-评论家算法在有效利用神经网络动力学的时候，处理了超出想象范围的奖励。为此，我们在学习到的潜在空间中预测状态价值和动作，如图1所示。这些价值函数针对想象奖励优化贝尔曼一致性，策略则通过将解析梯度反向传播至动力学模型来最大化这些价值函数。

与通过在线学习或经验回放（Lillicrap 等人, 2015; Mnih 等人, 2016; Schulman 等人, 2017; Haarnoja 等人, 2018; Lee 等人, 2019）进行学习的行者-评论家算法相比，世界模型能够插值过往经验，并针对多步回报提供解析梯度，从而实现高效的策略优化。

*通讯作者: Danijar Hafner <mail@danijar.com>。

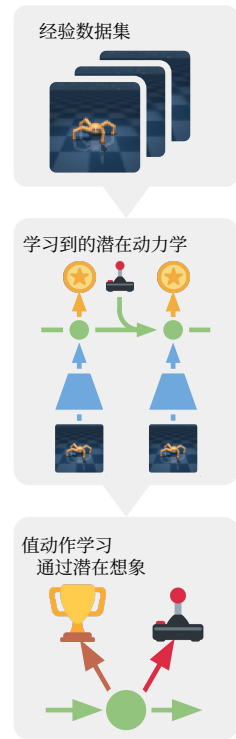


图1: Dreamer从过往经验中学习世界模型，并通过将价值估计反向传播回想象的轨迹，在其潜在空间中高效学习具有远见的行为。

DREAM TO CONTROL: LEARNING BEHAVIORS BY LATENT IMAGINATION

Danijar Hafner* Timothy Lillicrap Jimmy Ba Mohammad Norouzi
University of Toronto DeepMind University of Toronto Google Brain
Google Brain

Abstract

Learned world models summarize an agent’s experience to facilitate learning complex behaviors. While learning world models from high-dimensional sensory inputs is becoming feasible through deep learning, there are many potential ways for deriving behaviors from them. We present Dreamer, a reinforcement learning agent that solves long-horizon tasks from images purely by latent imagination. We efficiently learn behaviors by propagating analytic gradients of learned state values back through trajectories imagined in the compact state space of a learned world model. On 20 challenging visual control tasks, Dreamer exceeds existing approaches in data-efficiency, computation time, and final performance.

1 INTRODUCTION

Intelligent agents can achieve goals in complex environments even though they never encounter the exact same situation twice. This ability requires building representations of the world from past experience that enable generalization to novel situations. World models offer an explicit way to represent an agent’s knowledge about the world in a parametric model that can make predictions about the future.

When the sensory inputs are high-dimensional images, latent dynamics models can abstract observations to predict forward in compact state spaces (Watter et al., 2015; Oh et al., 2017; Gregor et al., 2019). Compared to predictions in image space, latent states have a small memory footprint that enables imagining thousands of trajectories in parallel. Learning effective latent dynamics models is becoming feasible through advances in deep learning and latent variable models (Krishnan et al., 2015; Karl et al., 2016; Doerr et al., 2018; Buesing et al., 2018).

Behaviors can be derived from dynamics models in many ways. Often, imagined rewards are maximized with a parametric policy (Sutton, 1991; Ha and Schmidhuber, 2018; Zhang et al., 2019) or by online planning (Chua et al., 2018; Hafner et al., 2018). However, considering only rewards within a fixed imagination horizon results in shortsighted behaviors (Wang et al., 2019). Moreover, prior work commonly resorts to derivative-free optimization for robustness to model errors (Ebert et al., 2017; Chua et al., 2018; Parmas et al., 2019), rather than leveraging analytic gradients offered by neural network dynamics (Henaff et al., 2019; Srinivas et al., 2018).

We present Dreamer, an agent that learns long-horizon behaviors from images purely by latent imagination. A novel actor critic algorithm accounts for rewards beyond the imagination horizon while making efficient use of the neural network dynamics. For this, we predict state values and actions in the learned latent space as summarized in Figure 1. The values optimize Bellman consistency for imagined rewards and the policy maximizes the values by propagating their analytic gradients back through the dynamics.

In comparison to actor critic algorithms that learn online or by experience replay (Lillicrap et al., 2015; Mnih et al., 2016; Schulman et al., 2017; Haarnoja et al., 2018; Lee et al., 2019), world models can interpolate past experience and offer analytic gradients of multi-step returns for efficient policy optimization.

*Correspondence to: Danijar Hafner <mail@danijar.com>.

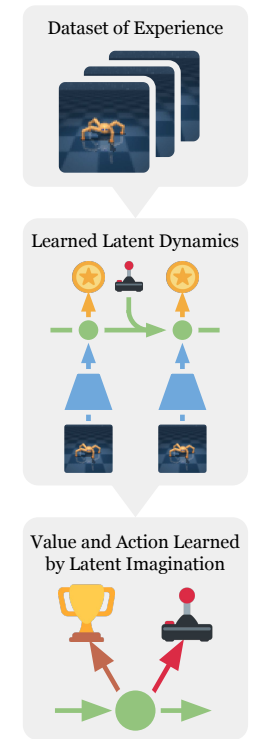


Figure 1: Dreamer learns a world model from past experience and efficiently learns farsighted behaviors in its latent space by backpropagating value estimates back through imagined trajectories.



图2: 我们实验中20个视觉控制任务中的5个的图像观测。这些任务提出了多种挑战, 包括接触动力学、稀疏奖励、多自由度以及三维环境。其中多个任务此前无法通过世界模型解决。

本文的主要贡献总结如下:

- **通过潜在想象学习长时域行为** 基于模型的智能体如果使用有限的想象范围, 可能会变得目光短浅。我们通过预测动作和状态价值来应对这一局限。在潜在空间中纯粹通过想象进行训练, 让我们能够通过将解析值梯度反向传播回潜在动力学, 从而高效地学习策略。
- **视觉控制的实证性能** 我们将Dreamer与现有的表征学习方法结合, 并在使用图像输入的DeepMind控制套件上进行评估, 如图2所示。对所有任务使用相同的超参数, Dreamer在数据效率、计算时间和最终性能方面均超越了以往的基于模型和无模型智能体。

2 使用世界模型进行控制

强化学习 我们将视觉控制建模为一个部分可观测马尔可夫决策过程 (POMDP), 具有离散时间步 $t \in [1; T]$, 由智能体生成的连续向量值动作 $a_t \sim p(a_t | p_{\leq t}, a_{< t})$, 以及由未知环境生成的高维观测和标量奖励 $o_t, r_t \sim p(o_t, r_t | o_{< t}, a_{< t})$ 。目标是开发一个能够最大化期望奖励总和 $E_p(\sum_{t=1}^T r_t)$ 的智能体。图2展示了我们选取的部分任务。

智能体组件 在想象中学习的智能体的经典组件包括动力学学习、行为学习和环境交互 (Sutton, 1991)。在Dreamer中, 行为是通过在世界模型的紧凑潜空间中预测假设轨迹来学习的。如图3所示并在算法1中详述, Dreamer在智能体的整个生命周期中执行以下操作, 这些操作可以交错进行或并行执行:

- 从包含过往经验的数据集中学习潜在动力学模型, 以根据动作和过去观测预测未来奖励。任何世界模型的学习目标都可以集成到Dreamer中。我们在第4节回顾了现有的学习潜在动力学方法。
- 根据预测的潜在轨迹学习动作模型和价值模型, 如第3节所述。价值模型优化想象奖励的贝尔曼一致性, 动作模型则通过将价值估计的梯度反向传播至神经网络动力学来更新。
- 在环境中执行学习到的动作模型, 以收集新的经验来扩充数据集。

潜在动力学 Dreamer 使用一个由三个组件组成的潜在动力学模型。表征模型编码观测和动作, 以创建具有马尔可夫转移的连续向量值模型状态 (s_t 参见 Watter等人, 2015; Zhang等人, 2019; Hafner等人, 2018)。转移模型预测未来的模型状态, 而无需看到随后会引发这些状态对应的观测。奖励模型根据模型状态预测奖励。

$$\begin{aligned}
 \text{表征模型:} & \quad p(s_t | s_{t-1}, a_{t-1}, o_t) \\
 \text{转移模型:} & \quad q(s_t | s_{t-1}, a_{t-1}) \\
 \text{奖励模型:} & \quad q(r_t | s_t).
 \end{aligned} \tag{1}$$

我们使用 p 表示在真实环境中生成样本的分布, 并使用 q 表示其近似分布, 从而实现潜在想象。具体而言, 转移模型使我们能够在紧凑潜空间中提前预测, 而无需观察或想象对应的图像。这使得内存占用低, 并能并行快速预测数千条想象轨迹。

该模型模拟了一种非线性卡尔曼滤波器 (卡尔曼, 1960)、潜在状态空间模型或具有实值状态的隐马尔可夫模型。然而, 它基于动作进行条件化并预测奖励, 从而允许智能体在不执行于环境中的情况下想象潜在动作序列的结果。



Figure 2: Image observations for 5 of the 20 visual control tasks used in our experiments. The tasks pose a variety of challenges including contact dynamics, sparse rewards, many degrees of freedom, and 3D environments. Several of these tasks could previously not be solved through world models.

The key contributions of this paper are summarized as follows:

- **Learning long-horizon behaviors by latent imagination** Model-based agents can be short-sighted if they use a finite imagination horizon. We approach this limitation by predicting both actions and state values. Training purely by imagination in a latent space lets us efficiently learn the policy by propagating analytic value gradients back through the latent dynamics.
- **Empirical performance for visual control** We pair Dreamer with existing representation learning methods and evaluate it on the DeepMind Control Suite with image inputs, illustrated in Figure 2. Using the same hyper parameters for all tasks, Dreamer exceeds previous model-based and model-free agents in terms of data-efficiency, computation time, and final performance.

2 CONTROL WITH WORLD MODELS

Reinforcement learning We formulate visual control as a partially observable Markov decision process (POMDP) with discrete time step $t \in [1; T]$, continuous vector-valued actions $a_t \sim p(a_t | o_{< t}, a_{< t})$ generated by the agent, and high-dimensional observations and scalar rewards $o_t, r_t \sim p(o_t, r_t | o_{< t}, a_{< t})$ generated by the unknown environment. The goal is to develop an agent that maximizes the expected sum of rewards $E_p(\sum_{t=1}^T r_t)$. Figure 2 shows a selection of our tasks.

Agent components The classical components of agents that learn in imagination are dynamics learning, behavior learning, and environment interaction (Sutton, 1991). In the case of Dreamer, the behavior is learned by predicting hypothetical trajectories in the compact latent space of the world model. As outlined in Figure 3 and detailed in Algorithm 1, Dreamer performs the following operations throughout the agent’s life time, either interleaved or in parallel:

- Learning the latent dynamics model from the dataset of past experience to predict future rewards from actions and past observations. Any learning objective for the world model can be incorporated with Dreamer. We review existing methods for learning latent dynamics in Section 4.
- Learning action and value models from predicted latent trajectories, as described in Section 3. The value model optimizes Bellman consistency for imagined rewards and the action model is updated by propagating gradients of value estimates back through the neural network dynamics.
- Executing the learned action model in the world to collect new experience for growing the dataset.

Latent dynamics Dreamer uses a latent dynamics model that consists of three components. The representation model encodes observations and actions to create continuous vector-valued model states s_t with Markovian transitions (Watter et al., 2015; Zhang et al., 2019; Hafner et al., 2018). The transition model predicts future model states without seeing the corresponding observations that will later cause them. The reward model predicts the rewards given the model states,

$$\begin{aligned}
 \text{Representation model:} & \quad p(s_t | s_{t-1}, a_{t-1}, o_t) \\
 \text{Transition model:} & \quad q(s_t | s_{t-1}, a_{t-1}) \\
 \text{Reward model:} & \quad q(r_t | s_t).
 \end{aligned} \tag{1}$$

We use p for distributions that generate samples in the real environment and q for their approximations that enable latent imagination. Specifically, the transition model lets us predict ahead in the compact latent space without having to observe or imagine the corresponding images. This results in a low memory footprint and fast predictions of thousands of imagined trajectories in parallel.

The model mimics a non-linear Kalman filter (Kalman, 1960), latent state space model, or HMM with real-valued states. However, it is conditioned on actions and predicts rewards, allowing the agent to imagine the outcomes of potential action sequences without executing them in the environment.

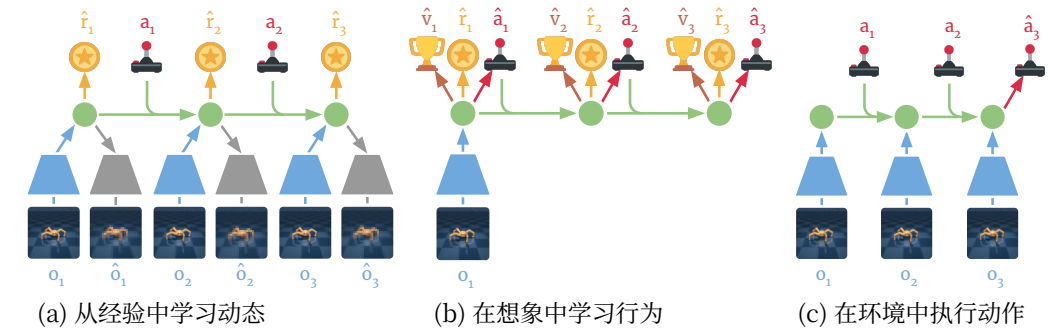


图3: Dreamer的组成部分。(a) 从过往经验数据集中, 智能体学习将观测和动作编码为紧凑潜在状态 (●), 例如通过重建, 并预测环境奖励 (●)。 (b) 在紧凑潜在空间中, Dreamer预测状态价值 (●) 和动作 (●), 通过沿想象轨迹传播梯度来最大化未来价值预测。(c) 智能体编码回合历史以计算当前模型状态, 并预测下一个要在环境中执行的动作。参见[算法1](#) 获取智能体的伪代码。

3 通过潜在想象学习行为

Dreamer 通过学习到的世界模型的紧凑潜空间中的长时域行为来学习, 通过高效利用神经网络潜在动力学。为此, 我们通过重参数化, 将多步回报的随机梯度通过动作、状态、奖励和价值的神经网络预测进行传播。本节描述了我们论文的主要贡献。

想象环境 潜在动力学定义了一个完全可观测的马尔可夫决策过程 (MDP; [Sutton,1991](#)), 因为紧凑的模型状态 s_t 是马尔可夫性的。我们用 τ 作为时间索引来标记想象量。想象轨迹从智能体过往经验中抽取的观测序列的真实模型状态 s_t 开始。它们遵循转移模型 $s_\tau \sim q(s_\tau | s_{\tau-1}, a_{\tau-1})$ 、奖励模型 $r_\tau \sim q(r_\tau | s_\tau)$ 和策略 $a_\tau \sim q(a_\tau | s_\tau)$ 的预测。目标是最大化关于策略的期望想象奖励 $E_q(\sum_{\tau=t}^{\infty} \gamma^{\tau-t} r_\tau)$ 。

算法1: Dreamer

初始化数据集 $\mathcal{D}$, 包含 S 个随机初始回合。随机初始化神经网络参数 θ, ϕ, ψ 。 **while** 未收敛 **do**

for 更新步骤 $c = 1..C$ **do**

// 动力学学习

抽取 B 数据序列 $\{(a_t, o_t, r_t)\}_{t=k}^{k+L} \sim \mathcal{D}$ 。

计算模型状态 $s_t \sim p_\theta(s_t | s_{t-1}, a_{t-1}, o_t)$ 。

使用表征学习更新 θ 。

// 行为学习

从每个 s_t 想象轨迹 $\{(s_\tau, a_\tau)\}_{\tau=t}^{t+H}$ 。

预测奖励 $E(q_\theta(r_\tau | s_\tau))$ 和价值函数 $v_\psi(s_\tau)$ 。

计算价值估计 $V_\lambda(s_\tau)$ 通过 [方程6](#)。

更新 $\phi \leftarrow \phi + \alpha \nabla_\phi \sum_{\tau=t}^{t+H} V_\lambda(s_\tau)$ 。更新

$\psi \leftarrow \psi - \alpha \nabla_\psi \sum_{\tau=t}^{t+H} \frac{1}{2} \|v_\psi(s_\tau) - V_\lambda(s_\tau)\|^2$ 。

// 环境交互

$o_1 \leftarrow \text{env.reset}()$

对于 时间步 $t = 1..T$ **执行**

从历史记录中计算 $s_t \sim p_\theta(s_t | s_{t-1}, a_{t-1}, o_t)$ 。

使用动作模型计算 $a_t \sim q_\phi(a_t | s_t)$ 。

向动作中添加探索噪声。

$r_t, o_{t+1} \leftarrow \text{env.step}(a_t)$ 。

将经验添加到数据集 $\mathcal{D} \leftarrow \mathcal{D} \cup \{(o_t, a_t, r_t)_{t=1}^T\}$ 。

模型组件

表示

$$p_\theta(s_t | s_{t-1}, a_{t-1}, o_t)$$

转移

$$q_\theta(s_t | s_{t-1}, a_{t-1})$$

奖励

$$q_\theta(r_t | s_t)$$

动作

$$q_\phi(a_t | s_t)$$

值

$$v_\psi(s_t)$$

超参数

初始回合

S

收集间隔

C

批量大小

B

序列长度

L

想象范围

H

学习率

α

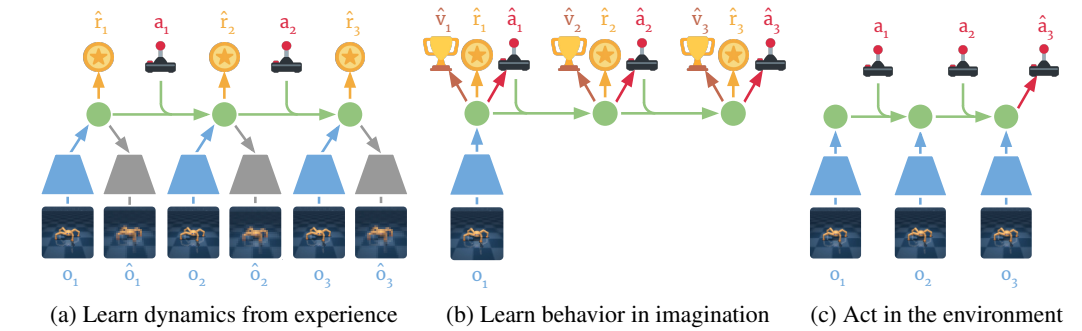


Figure 3: Components of Dreamer. (a) From the dataset of past experience, the agent learns to encode observations and actions into compact latent states (●), for example via reconstruction, and predicts environment rewards (●). (b) In the compact latent space, Dreamer predicts state values (●) and actions (●) that maximize future value predictions by propagating gradients back through imagined trajectories. (c) The agent encodes the history of the episode to compute the current model state and predict the next action to execute in the environment. See [Algorithm 1](#) for pseudo code of the agent.

3 LEARNING BEHAVIORS BY LATENT IMAGINATION

Dreamer learns long-horizon behaviors in the compact latent space of a learned world model by efficiently leveraging the neural network latent dynamics. For this, we propagate stochastic gradients of multi-step returns through neural network predictions of actions, states, rewards, and values using reparameterization. This section describes the main contribution of our paper.

Imagination environment The latent dynamics define a Markov decision process (MDP; [Sutton, 1991](#)) that is fully observed because the compact model states s_t are Markovian. We denote imagined quantities with τ as the time index. Imagined trajectories start at the true model states s_t of observation sequences drawn from the agent’s past experience. They follow predictions of the transition model $s_\tau \sim q(s_\tau | s_{\tau-1}, a_{\tau-1})$, reward model $r_\tau \sim q(r_\tau | s_\tau)$, and a policy $a_\tau \sim q(a_\tau | s_\tau)$. The objective is to maximize expected imagined rewards $E_q(\sum_{\tau=t}^{\infty} \gamma^{\tau-t} r_\tau)$ with respect to the policy.

Algorithm 1: Dreamer

Initialize dataset $\mathcal{D}$ with S random seed episodes.

Initialize neural network parameters θ, ϕ, ψ randomly.

while not converged do

for update step $c = 1..C$ **do**

// Dynamics learning

Draw B data sequences $\{(a_t, o_t, r_t)\}_{t=k}^{k+L} \sim \mathcal{D}$.

Compute model states $s_t \sim p_\theta(s_t | s_{t-1}, a_{t-1}, o_t)$.

Update θ using representation learning.

// Behavior learning

Imagine trajectories $\{(s_\tau, a_\tau)\}_{\tau=t}^{t+H}$ from each s_t .

Predict rewards $E(q_\theta(r_\tau | s_\tau))$ and values $v_\psi(s_\tau)$.

Compute value estimates $V_\lambda(s_\tau)$ via [Equation 6](#).

Update $\phi \leftarrow \phi + \alpha \nabla_\phi \sum_{\tau=t}^{t+H} V_\lambda(s_\tau)$.

Update $\psi \leftarrow \psi - \alpha \nabla_\psi \sum_{\tau=t}^{t+H} \frac{1}{2} \|v_\psi(s_\tau) - V_\lambda(s_\tau)\|^2$.

// Environment interaction

$o_1 \leftarrow \text{env.reset}()$

for time step $t = 1..T$ **do**

Compute $s_t \sim p_\theta(s_t | s_{t-1}, a_{t-1}, o_t)$ from history.

Compute $a_t \sim q_\phi(a_t | s_t)$ with the action model.

Add exploration noise to action.

$r_t, o_{t+1} \leftarrow \text{env.step}(a_t)$.

Add experience to dataset $\mathcal{D} \leftarrow \mathcal{D} \cup \{(o_t, a_t, r_t)_{t=1}^T\}$.

Model components

Representation $p_\theta(s_t | s_{t-1}, a_{t-1}, o_t)$

Transition $q_\theta(s_t | s_{t-1}, a_{t-1})$

Reward $q_\theta(r_t | s_t)$

Action $q_\phi(a_t | s_t)$

Value $v_\psi(s_t)$

Hyper parameters

Seed episodes S

Collect interval C

Batch size B

Sequence length L

Imagination horizon H

Learning rate α

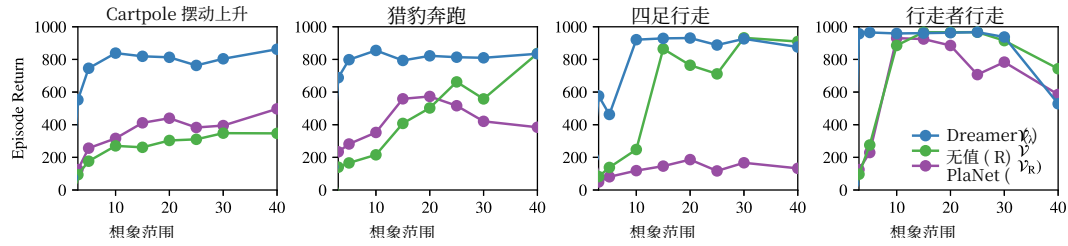


图 4: 想象范围。我们比较了 Dreamer 的最终性能、没有价值预测的动作模型学习以及使用 PlaNet 的在线规划。学习一个状态价值模型来估计超出想象范围的奖励, 使得 Dreamer 对想象范围的长度更加鲁棒。智能体使用像素重建进行表征学习, 并采用 $R = 2$ 的动作重复。

动作模型和价值模型 考虑有限视野 H 内的想象轨迹。Dreamer 采用演员-评论家方法学习行为, 以考虑超出视野的奖励。为此, 我们在世界模型的潜在空间中学习动作模型和价值模型。动作模型实现了策略, 旨在预测能够解决想象环境的动作。价值模型估计动作模型从每个状态 s_τ 所能达到的期望想象奖励,

$$\begin{aligned} \text{动作模型:} \quad & a_\tau \sim q_\phi(a_\tau | s_\tau) \\ \text{价值模型:} \quad & v_\psi(s_\tau) \approx \mathbb{E}_{q(\cdot|s_\tau)} \left(\sum_{\tau=t}^{t+H} \gamma^{\tau-t} r_\tau \right). \end{aligned} \quad (2)$$

动作模型和价值模型按照策略迭代中的典型方式协同训练: 动作模型旨在最大化价值的估计值, 而价值模型旨在匹配一个随着动作模型变化而变化的估计值。

我们使用密集神经网络分别构建动作模型和价值模型, 其参数为 ϕ 和 ψ 。动作模型输出一个双曲正切变换高斯分布 (Haarnoja 等人, 2018), 其充分统计量由神经网络预测。这使得可以采用重参数化采样 (Kingma 和 Welling, 2013; Rezende 等人, 2014), 将采样得到的动作视为神经网络输出的确定性函数, 从而允许我们通过采样操作反向传播解析梯度。

$$a_\tau = \tanh(\mu_\phi(s_\tau) + \sigma_\phi(s_\tau)\epsilon), \quad \epsilon \sim \text{Normal}(0, \mathbb{I}). \quad (3)$$

价值估计 为了学习动作模型和价值模型, 我们需要估计想象轨迹的状态价值 $\{s_\tau, a_\tau, r_\tau\}_{\tau=t}^{t+H}$ 。这些轨迹从模型状态 s_t 分支出来, 这些模型状态来自从智能体的经验数据集中抽取的序列批次, 并利用从动作模型中采样的动作向前预测想象范围 H 。状态价值可以通过多种方式进行估计, 这些方式在偏差与方差之间进行权衡 (Sutton 和 Barto, 2018),

$$V_R(s_\tau) \doteq \mathbb{E}_{q_\theta, q_\phi} \left(\sum_{n=\tau}^{t+H} r_n \right), \quad (4)$$

$$V_N^k(s_\tau) \doteq \mathbb{E}_{q_\theta, q_\phi} \left(\sum_{n=\tau}^{h-1} \gamma^{n-\tau} r_n + \gamma^{h-\tau} v_\psi(s_h) \right) \quad \text{with} \quad h = \min(\tau + k, t + H), \quad (5)$$

$$V_\lambda(s_\tau) \doteq (1 - \lambda) \sum_{n=1}^{H-1} \lambda^{n-1} V_N^n(s_\tau) + \lambda^{H-1} V_N^H(s_\tau), \quad (6)$$

其中期望值是在想象轨迹下估计的。 V_R 简单地将从 τ 到想象范围的奖励累加, 并忽略该范围之外的奖励。这使得可以在没有价值模型的情况下学习动作模型, 我们在实验中将其作为消融对比。 V_N^k 使用学到的价值模型估计 k 步之后的奖励。Dreamer 使用 V_λ , 即对不同 k 的估计值进行指数加权平均, 以平衡偏差与方差。图4表明, 在想象中学习价值模型使 Dreamer 能够解决长时域任务, 同时对想象范围具有鲁棒性。所有任务的实验细节和结果详见第6节。

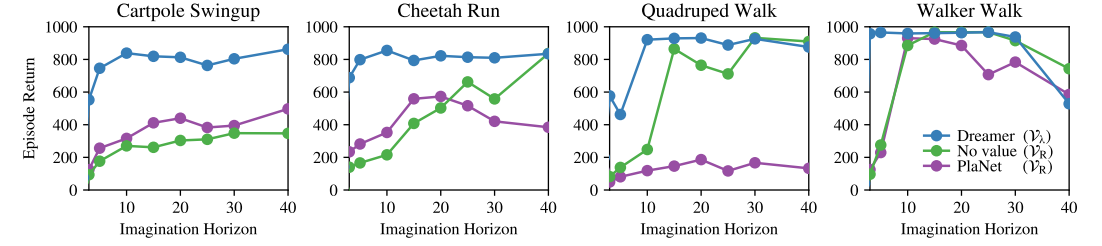


Figure 4: Imagination horizons. We compare the final performance of Dreamer, learning an action model without value prediction, and online planning using PlaNet. Learning a state value model to estimate rewards beyond the imagination horizon makes Dreamer more robust to the horizon length. The agents use pixel reconstruction for representation learning and an action repeat of $R = 2$.

Action and value models Consider imagined trajectories with a finite horizon H . Dreamer uses an actor critic approach to learn behaviors that consider rewards beyond the horizon. We learn an action model and a value model in the latent space of the world model for this. The action model implements the policy and aims to predict actions that solve the imagination environment. The value model estimates the expected imagined rewards that the action model achieves from each state s_τ ,

$$\begin{aligned} \text{Action model:} \quad & a_\tau \sim q_\phi(a_\tau | s_\tau) \\ \text{Value model:} \quad & v_\psi(s_\tau) \approx \mathbb{E}_{q(\cdot|s_\tau)} \left(\sum_{\tau=t}^{t+H} \gamma^{\tau-t} r_\tau \right). \end{aligned} \quad (2)$$

The action and value models are trained cooperatively as typical in policy iteration: the action model aims to maximize an estimate of the value, while the value model aims to match an estimate of the value that changes as the action model changes.

We use dense neural networks for the action and value models with parameters ϕ and ψ , respectively. The action model outputs a tanh-transformed Gaussian (Haarnoja et al., 2018) with sufficient statistics predicted by the neural network. This allows for reparameterized sampling (Kingma and Welling, 2013; Rezende et al., 2014) that views sampled actions as deterministically dependent on the neural network output, allowing us to backpropagate analytic gradients through the sampling operation,

$$a_\tau = \tanh(\mu_\phi(s_\tau) + \sigma_\phi(s_\tau)\epsilon), \quad \epsilon \sim \text{Normal}(0, \mathbb{I}). \quad (3)$$

Value estimation To learn the action and value models, we need to estimate the state values of imagined trajectories $\{s_\tau, a_\tau, r_\tau\}_{\tau=t}^{t+H}$. These trajectories branch off of the model states s_t of sequence batches drawn from the agent’s dataset of experience and predict forward for the imagination horizon H using actions sampled from the action model. State values can be estimated in multiple ways that trade off bias and variance (Sutton and Barto, 2018),

$$V_R(s_\tau) \doteq \mathbb{E}_{q_\theta, q_\phi} \left(\sum_{n=\tau}^{t+H} r_n \right), \quad (4)$$

$$V_N^k(s_\tau) \doteq \mathbb{E}_{q_\theta, q_\phi} \left(\sum_{n=\tau}^{h-1} \gamma^{n-\tau} r_n + \gamma^{h-\tau} v_\psi(s_h) \right) \quad \text{with} \quad h = \min(\tau + k, t + H), \quad (5)$$

$$V_\lambda(s_\tau) \doteq (1 - \lambda) \sum_{n=1}^{H-1} \lambda^{n-1} V_N^n(s_\tau) + \lambda^{H-1} V_N^H(s_\tau), \quad (6)$$

where the expectations are estimated under the imagined trajectories. V_R simply sums the rewards from τ until the horizon and ignores rewards beyond it. This allows learning the action model without a value model, an ablation we compare to in our experiments. V_N^k estimates rewards beyond k steps with the learned value model. Dreamer uses V_λ , an exponentially-weighted average of the estimates for different k to balance bias and variance. Figure 4 shows that learning a value model in imagination enables Dreamer to solve long-horizon tasks while being robust to the imagination horizon. The experimental details and results on all tasks are described in Section 6.

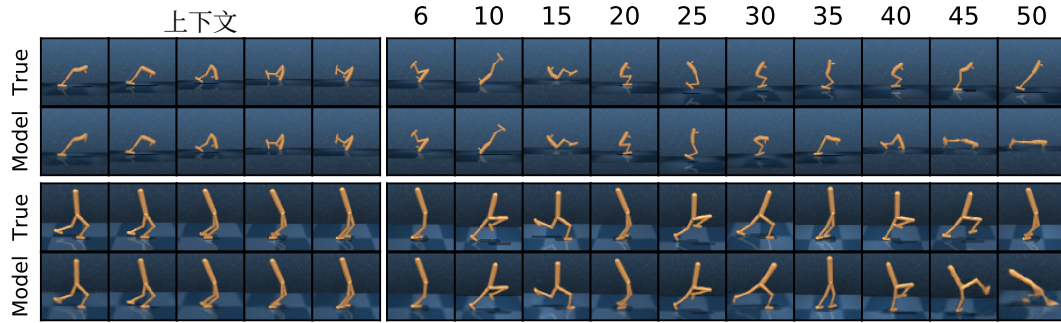


图5：长期预测的重建效果。我们将表征模型应用于两个保留轨迹的前5帧图像，并仅根据动作，利用潜在动力学向前预测45步。循环状态空间模型（RSSM；Hafner 等人, 2018）能够进行准确的长期预测，从而使 Dreamer 能够在紧凑潜空间中学习成功的行为。

学习目标 为了更新动作模型和价值模型，我们首先计算沿想象轨迹的所有状态 s_τ 的价值估计值 $V_\lambda(s_\tau)$ 。动作模型 $q_\phi(a_\tau | s_\tau)$ 的目标是预测能够产生高价值估计值状态轨迹的动作。而价值模型 $v_\psi(s_\tau)$ 的目标则是回归这些价值估计值，

$$\max_{\phi} \mathbb{E}_{q_\theta, q_\phi} \left(\sum_{\tau=t}^{t+H} V_\lambda(s_\tau) \right), \quad (7) \quad \min_{\psi} \mathbb{E}_{q_\theta, q_\phi} \left(\sum_{\tau=t}^{t+H} \frac{1}{2} \left\| v_\psi(s_\tau) - V_\lambda(s_\tau) \right\|^2 \right). \quad (8)$$

价值模型通过回归目标值进行更新，在计算目标值时，我们按照惯例（Sutton 和 Barto, 2018）对其停止梯度计算。动作模型则利用通过已学习动力学计算得到的解析梯度来最大化价值估计。理解这一点需要注意：价值估计依赖于奖励预测和价值预测，而奖励预测和价值预测又依赖于想象的状态，这些状态进而依赖于想象的动作。由于所有步骤均通过神经网络实现，我们通过随机反向传播（Kingma 和 Welling, 2013; Rezende 等人, 2014）解析地计算 $\nabla_{\phi} \mathbb{E}_{q_\theta, q_\phi} \left(\sum_{\tau=t}^{t+H} V_\lambda(s_\tau) \right)$ 。对于连续动作和潜在状态，我们使用重参数化；对于离散动作，则使用直通梯度（Bengio 等人, 2013）。在学习行为的过程中，世界模型保持固定。在具有提前终止的任务中，世界模型还会从每个潜在状态预测折扣因子，以根据预测折扣因子的累积乘积对公式 7 和 8 中的时间步进行加权，从而根据想象轨迹可能提前终止的可能性来降低各项的权重。与演员-评论家方法的比较：使用 REINFORCE 梯度（Williams, 1992）的智能体（例如 A3C 和 PPO（Mnih 等人, 2016; Schulman 等人, 2017））采用价值基线来减少梯度方差，而 Dreamer 则通过价值模型进行反向传播。这与确定性或重参数化演员-评论家方法（Silver 等人, 2014）（例如 DDPG 和 SAC（Lillicrap 等人, 2015; Haarnoja 等人, 2018））类似。然而，这些方法并未利用通过转移计算的梯度，并且仅最大化即时 Q 值。MVE 和 STEVE（Feinberg 等人, 2018; Buckman 等人, 2018）将其扩展到基于已学习动力学的多步 Q 学习，以提供更准确的 Q 值目标。我们预测状态价值，由于可以通过动力学进行反向传播，这对于策略优化而言已经足够。更详细的与相关工作的比较，请参见第 5 节。

4 学习潜在动力学

在想象中学习行为需要一个泛化能力强的世界模型。我们重点关注在紧凑潜空间中进行前向预测的潜在动力学模型，这有助于进行长期预测，并允许智能体并行想象数千条轨迹。目前已有多种用于学习控制相关表征的目标函数被提出（Watter 等人, 2015; Jaderberg 等人, 2016; Oord 等人, 2018; Eslami 等人, 2018）。我们回顾了三种用于 Dreamer 的表征学习方法：奖励预测、图像重建和对比估计。

奖励预测 潜在想象需要一个表征模型 $p(s_t | s_{t-1}, a_{t-1}, o_t)$ 、转移模型 $q(s_t | s_{t-1}, a_{t-1},)$ 和奖励模型 $q(r_t | s_t)$ ，如第 2 节所述。原则上，这可以通过简单地学习根据动作和过去观测来预测未来奖励来实现（Oh 等人, 2017; Gelada 等人, 2019; Schrittwieser 等人, 2019）。拥有大规模多样化数据集时，这样的表示应足以解决控制任务。然而，在有限数据集且尤其是奖励稀疏的情况下，学习与奖励相关的观测可能会改进世界模型（Jaderberg 等人, 2016; Gregor 等人, 2019）。

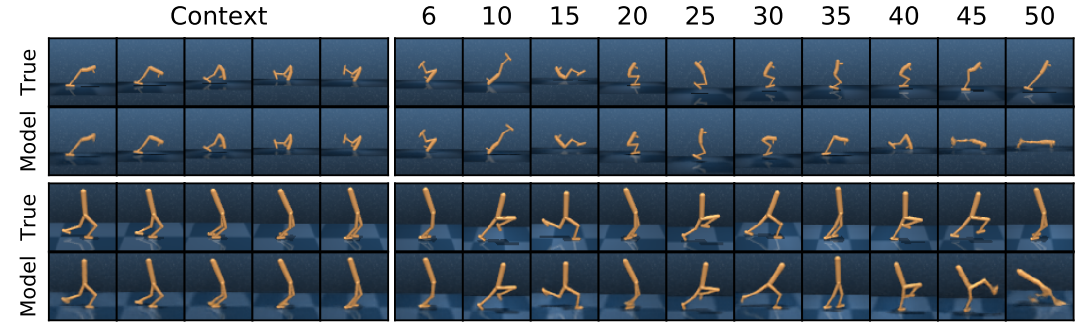


Figure 5: Reconstructions of long-term predictions. We apply the representation model to the first 5 images of two hold-out trajectories and predict forward for 45 steps using the latent dynamics, given only the actions. The recurrent state space model (RSSM; Hafner et al., 2018) performs accurate long-term predictions, enabling Dreamer to learn successful behaviors in a compact latent space.

Learning objective To update the action and value models, we first compute the value estimates $V_\lambda(s_\tau)$ for all states s_τ along the imagined trajectories. The objective for the action model $q_\phi(a_\tau | s_\tau)$ is to predict actions that result in state trajectories with high value estimates. The objective for the value model $v_\psi(s_\tau)$, in turn, is to regress the value estimates,

$$\max_{\phi} \mathbb{E}_{q_\theta, q_\phi} \left(\sum_{\tau=t}^{t+H} V_\lambda(s_\tau) \right), \quad (7) \quad \min_{\psi} \mathbb{E}_{q_\theta, q_\phi} \left(\sum_{\tau=t}^{t+H} \frac{1}{2} \left\| v_\psi(s_\tau) - V_\lambda(s_\tau) \right\|^2 \right). \quad (8)$$

The value model is updated to regress the targets, around which we stop the gradient as typical (Sutton and Barto, 2018). The action model uses analytic gradients through the learned dynamics to maximize the value estimates. To understand this, we note that the value estimates depend on the reward and value predictions, which depend on the imagined states, which in turn depend on the imagined actions. Since all steps are implemented as neural networks, we analytically compute $\nabla_{\phi} \mathbb{E}_{q_\theta, q_\phi} \left(\sum_{\tau=t}^{t+H} V_\lambda(s_\tau) \right)$ by stochastic backpropagation (Kingma and Welling, 2013; Rezende et al., 2014). We use reparameterization for continuous actions and latent states and straight-through gradients (Bengio et al., 2013) for discrete actions. The world model is fixed while learning behaviors.

In tasks with early termination, the world model also predicts the discount factor from each latent state to weigh the time steps in Equations 7 and 8 by the cumulative product of the predicted discount factors, so terms are weighted down based on how likely the imagined trajectory would have ended.

Comparison to actor critic methods Agents using Reinforce gradients (Williams, 1992), such as A3C and PPO (Mnih et al., 2016; Schulman et al., 2017), employ value baselines to reduce gradient variance, while Dreamer backpropagates through the value model. This is similar to deterministic or reparameterized actor critics (Silver et al., 2014), such as DDPG and SAC (Lillicrap et al., 2015; Haarnoja et al., 2018). However, these do not leverage gradients through transitions and only maximize immediate Q-values. MVE and STEVE (Feinberg et al., 2018; Buckman et al., 2018) extend them to multi-step Q-learning with learned dynamics to provide more accurate Q-value targets. We predict state values, which is sufficient for policy optimization since we backpropagate through the dynamics. Refer to Section 5 for a more detailed comparison to related work.

4 LEARNING LATENT DYNAMICS

Learning behaviors in imagination requires a world model that generalizes well. We focus on latent dynamics models that predict forward in a compact latent space, facilitating long-term predictions and allowing the agent to imagine thousands of trajectories in parallel. Several objectives for learning representations for control have been proposed (Watter et al., 2015; Jaderberg et al., 2016; Oord et al., 2018; Eslami et al., 2018). We review three approaches for learning representations to use with Dreamer: reward prediction, image reconstruction, and contrastive estimation.

Reward prediction Latent imagination requires a representation model $p(s_t | s_{t-1}, a_{t-1}, o_t)$, transition model $q(s_t | s_{t-1}, a_{t-1},)$, and reward model $q(r_t | s_t)$, as described in Section 2. In principle, this could be achieved by simply learning to predict future rewards given actions and past observations (Oh et al., 2017; Gelada et al., 2019; Schrittwieser et al., 2019). With a large and diverse dataset, such representations should be sufficient for solving a control task. However, with a finite dataset and especially when rewards are sparse, learning about observations that correlate with rewards is likely to improve the world model (Jaderberg et al., 2016; Gregor et al., 2019).

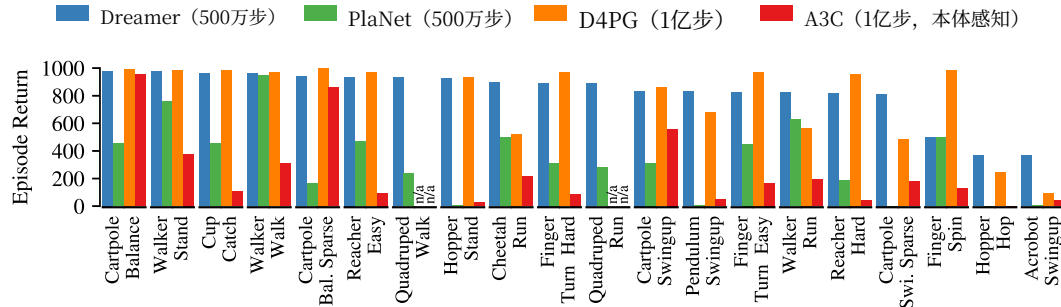


图6: 各方法性能对比。Dreamer继承了PlaNet的数据效率, 同时超越了最优无模型智能体的渐进性能。在 5×10^6 环境步数后, Dreamer跨任务平均性能达到823, 而PlaNet为332, 最优无模型D4PG智能体在 10^8 步后达到786。结果为5个随机种子的平均值。

重建 我们首先描述PlaNet使用的世界模型 (Hafner 等人, 2018), 该模型通过重建图像来学习潜在动力学, 如图3a所示。该世界模型由以下组件构成, 其中观测模型仅用于提供学习信号,

$$\begin{aligned}
 \text{表征模型:} & p_{\theta}(s_t | s_{t-1}, a_{t-1}, o_t) \\
 \text{观测模型:} & q_{\theta}(o_t | s_t) \\
 \text{奖励模型:} & q_{\theta}(r_t | s_t) \\
 \text{转移模型:} & q_{\theta}(s_t | s_{t-1}, a_{t-1}).
 \end{aligned} \tag{9}$$

这些组件被联合优化以提高变分下界 (ELBO; Jordan 等人, 1999), 或更一般地, 变分信息瓶颈 (VIB; Tishby 等人, 2000; Alemi 等人, 2016)。如附录B所推导, 该界包括观测和奖励的重构项以及一个KL正则化项。该期望是在数据集和表征模型下计算的。

$$\begin{aligned}
 \mathcal{J}_{\text{REC}} & \doteq \mathbb{E}_p \left(\sum_t \left(\mathcal{J}_{\text{O}}^t + \mathcal{J}_{\text{R}}^t + \mathcal{J}_{\text{D}}^t \right) \right) + \text{const} & \mathcal{J}_{\text{O}}^t & \doteq \ln q(o_t | s_t) \\
 \mathcal{J}_{\text{R}}^t & \doteq \ln q(r_t | s_t) & \mathcal{J}_{\text{D}}^t & \doteq -\beta \text{KL} \left(p(s_t | s_{t-1}, a_{t-1}, o_t) \parallel q(s_t | s_{t-1}, a_{t-1}) \right).
 \end{aligned} \tag{10}$$

我们将转移模型实现为循环状态空间模型 (RSSM; Hafner 等人, 2018), 表征模型则通过将RSSM与应用于图像观测的卷积神经网络 (CNN; LeCun等人, 1989) 相结合来实现, 观测模型实现为转置CNN, 奖励模型实现为密集网络。组合参数向量 θ 通过随机反向传播进行更新 (Kingma和Welling, 2013; Rezende等人, 2014)。图5展示了该模型的视频预测结果。关于模型细节, 我们参考附录A和Hafner 等人(2018)。

对比估计 预测像素可能需要较高的模型容量。我们还可以通过从图像中预测状态来鼓励模型状态与观测之间的互信息 (Guo等人, 2018)。这将以状态模型替换观测模型,

$$\text{State model:} \quad q_{\theta}(s_t | o_t). \tag{11}$$

重建目标利用了观测边缘分布是常数这一事实, 而现在我们面临的是状态边缘分布。如附录B所示, 这可以通过对当前序列批次中的观测进行状态模型平均, 利用噪声对比估计 (NCE; Gutmann和Hyvärinen, 2010; Oord等人, 2018) 来估计。直观上, $q(s_t | o_t)$ 使得状态能从当前图像中可预测, 而 $\ln \sum_{o'} q(s_t | o')$ 则保持其多样性以防止崩溃,

$$\mathcal{J}_{\text{NCE}} \doteq \mathbb{E} \left(\sum_t \left(\mathcal{J}_{\text{S}}^t + \mathcal{J}_{\text{R}}^t + \mathcal{J}_{\text{D}}^t \right) \right) \quad \mathcal{J}_{\text{S}}^t \doteq \ln q(s_t | o_t) - \ln \left(\sum_{o'} q(s_t | o') \right). \tag{12}$$

我们将状态模型实现为CNN, 并再次相对于组合参数向量优化该界 θ , 使用随机反向传播。虽然避免了像素预测, 但该界能够高效提取的信息量是有限的 (McAllester 和 Statos, 2018)。我们在实验中对奖励、重建和对比目标进行了实证比较, 详见图8。

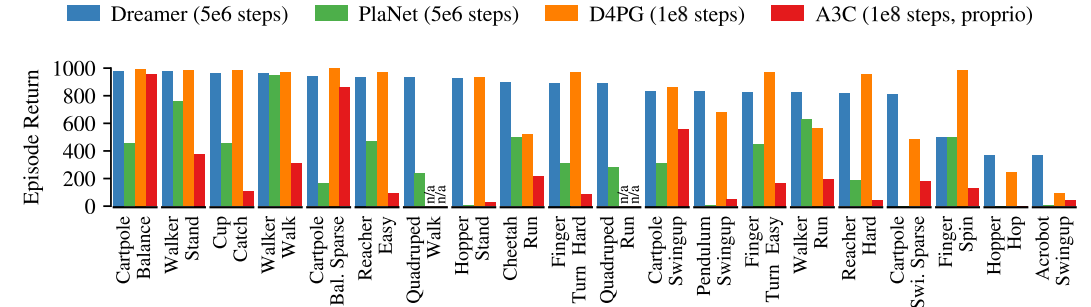


Figure 6: Performance comparison to existing methods. Dreamer inherits the data-efficiency of PlaNet while exceeding the asymptotic performance of the best model-free agents. After 5×10^6 environment steps, Dreamer reaches an average performance of 823 across tasks, compared to PlaNet at 332 and the top model-free D4PG agent at 786 after 10^8 steps. Results are averages over 5 seeds.

Reconstruction We first describe the world model used by PlaNet (Hafner et al., 2018) that learns latent dynamics by reconstructing images as shown in Figure 3a. The world model consists of the following components, where the observation model is only used to provide a learning signal,

$$\begin{aligned}
 \text{Representation model:} & p_{\theta}(s_t | s_{t-1}, a_{t-1}, o_t) \\
 \text{Observation model:} & q_{\theta}(o_t | s_t) \\
 \text{Reward model:} & q_{\theta}(r_t | s_t) \\
 \text{Transition model:} & q_{\theta}(s_t | s_{t-1}, a_{t-1}).
 \end{aligned} \tag{9}$$

The components are optimized jointly to increase the variational lower bound (ELBO; Jordan et al., 1999) or more generally the variational information bottleneck (VIB; Tishby et al., 2000; Alemi et al., 2016). As derived in Appendix B, the bound includes reconstruction terms for observations and rewards and a KL regularizer. The expectation is taken under the dataset and representation model,

$$\begin{aligned}
 \mathcal{J}_{\text{REC}} & \doteq \mathbb{E}_p \left(\sum_t \left(\mathcal{J}_{\text{O}}^t + \mathcal{J}_{\text{R}}^t + \mathcal{J}_{\text{D}}^t \right) \right) + \text{const} & \mathcal{J}_{\text{O}}^t & \doteq \ln q(o_t | s_t) \\
 \mathcal{J}_{\text{R}}^t & \doteq \ln q(r_t | s_t) & \mathcal{J}_{\text{D}}^t & \doteq -\beta \text{KL} \left(p(s_t | s_{t-1}, a_{t-1}, o_t) \parallel q(s_t | s_{t-1}, a_{t-1}) \right).
 \end{aligned} \tag{10}$$

We implement the transition model as a recurrent state space model (RSSM; Hafner et al., 2018), the representation model by combining the RSSM with a convolutional neural network (CNN; LeCun et al., 1989) applied to the image observation, the observation model as a transposed CNN, and the reward model as a dense network. The combined parameter vector θ is updated by stochastic backpropagation (Kingma and Welling, 2013; Rezende et al., 2014). Figure 5 shows video predictions of this model. We refer to Appendix A and Hafner et al. (2018) model details.

Contrastive estimation Predicting pixels can require high model capacity. We can also encourage mutual information between model states and observations by instead predicting the states from the images (Guo et al., 2018). This replaces the observation model with a state model,

$$\text{State model:} \quad q_{\theta}(s_t | o_t). \tag{11}$$

While the reconstruction objective used the fact that the observation marginal is a constant, we now face the state marginal. As shown in Appendix B, this can be estimated via noise contrastive estimation (NCE; Gutmann and Hyvärinen, 2010; Oord et al., 2018) by averaging the state model over observations o' of the current sequence batch. Intuitively, $q(s_t | o_t)$ makes the state predictable from the current image while $\ln \sum_{o'} q(s_t | o')$ keeps it diverse to prevent collapse,

$$\mathcal{J}_{\text{NCE}} \doteq \mathbb{E} \left(\sum_t \left(\mathcal{J}_{\text{S}}^t + \mathcal{J}_{\text{R}}^t + \mathcal{J}_{\text{D}}^t \right) \right) \quad \mathcal{J}_{\text{S}}^t \doteq \ln q(s_t | o_t) - \ln \left(\sum_{o'} q(s_t | o') \right). \tag{12}$$

We implement the state model as a CNN and again optimize the bound with respect to the combined parameter vector θ using stochastic backpropagation. While avoiding pixel prediction, the amount of information this bound can extract efficiently is limited (McAllester and Statos, 2018). We empirically compare reward, reconstruction, and contrastive objectives in our experiments in Figure 8.

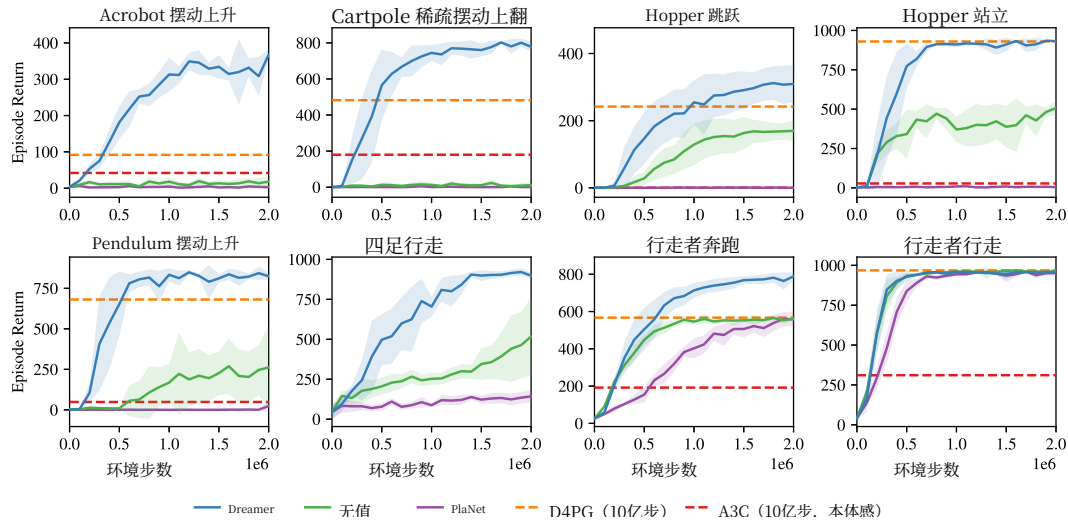


图 7: Dreamer 在需要长程信用分配的视觉控制任务（如 Acrobot 和 Hopper 任务）中取得成功。通过动作模型或在线规划仅优化视界内的想象奖励，会产生短视行为，仅在反应型任务（如行走者领域）中取得成功。所有 20 个任务的性能总结见 图 6，训练曲线见附录 D。有关 D4PG 和 A3C 的性能曲线，请参阅Tassa 等人(2018)。

5 相关工作

先前的工作通过无导数策略学习或在线规划来学习用于视觉控制的潜在动力学，用多步预测增强无模型智能体，或使用Q值或多步奖励的解析梯度，这通常适用于低维任务。相比之下，Dreamer 纯粹通过潜在想象，利用解析梯度高效地学习用于视觉控制的长时域行为。

基于潜在动力学的控制 E2C (Watter 等人, 2015) 和 RCE (Banijamali 等人, 2017) 将图像嵌入紧凑空间进行前向预测，以解决简单任务。World Models (Ha 和 Schmidhuber, 2018) 通过两阶段过程学习潜在动力学，以便在想象中演化线性控制器。PlaNet (Hafner 等人, 2018) 联合学习潜在动力学，并通过潜在在线规划解决视觉移动任务。SOLAR (Zhang 等人, 2019) 通过在潜在空间中进行引导策略搜索来解决机器人任务。I2A (Weber 等人, 2017) 将想象轨迹交给无模型策略，而Lee 等人(2019) 和Gregor 等人(2019) 学习信念表示以加速无模型智能体。

想象多步回报VPN (Oh等人, 2017)、MVE (Feinberg等人, 2018) 和 STEVE (Buckman等人, 2018) 从回放缓冲器中学习动力学以进行多步Q学习。AlphaGo (Silver等人, 2017) 将动作和状态价值的预测与规划相结合，并假设能访问真实动力学。同样假设能访问动力学，POLO (Lowrey等人, 2018) 通过学习价值集成来规划探索。MuZero (Schrittwieser等人, 2019) 学习任务特定奖励和价值模型以解决具有挑战性的任务，但需要大量经验。PETS (Chua 等人, 2018)、VisualMPC (Ebert 等人, 2017) 和 PlaNet (Hafner等人, 2018) 使用无导数优化进行在线规划。POPLIN (Wang和Ba, 2019) 通过自我模仿改进了在线规划。Piergiovanni等人 (2018) 通过潜在动力学模型的想象学习机器人策略。基于神经网络梯度的规划已在小型问题上得到展示 (Schmidhuber, 1990; Henaff 等人, 2018)，但规模化一直具有挑战性 (Parmas 等人, 2019)。

解析值梯度 DPG (Silver 等人, 2014)、DDPG (Lillicrap 等人, 2015) 和 SAC (Haarnoja 等人, 2018) 利用习得的即时动作值的梯度，通过经验回放来学习策略。SVG (Heess 等人, 2015) 通过单步模型预测的解析值梯度，降低了无模型同策略算法的方差。Byravan 等人 (2019) 的同期工作将确定性模型的潜在想象用于导航和操作任务。ME-TRPO (Kurutach 等人, 2018) 通过本体感知输入的预测奖励梯度，加速了原本无模型的智能体。DistGBP (Henaff 等人, 2017; 2019) 在简单任务中使用模型梯度进行在线规划。

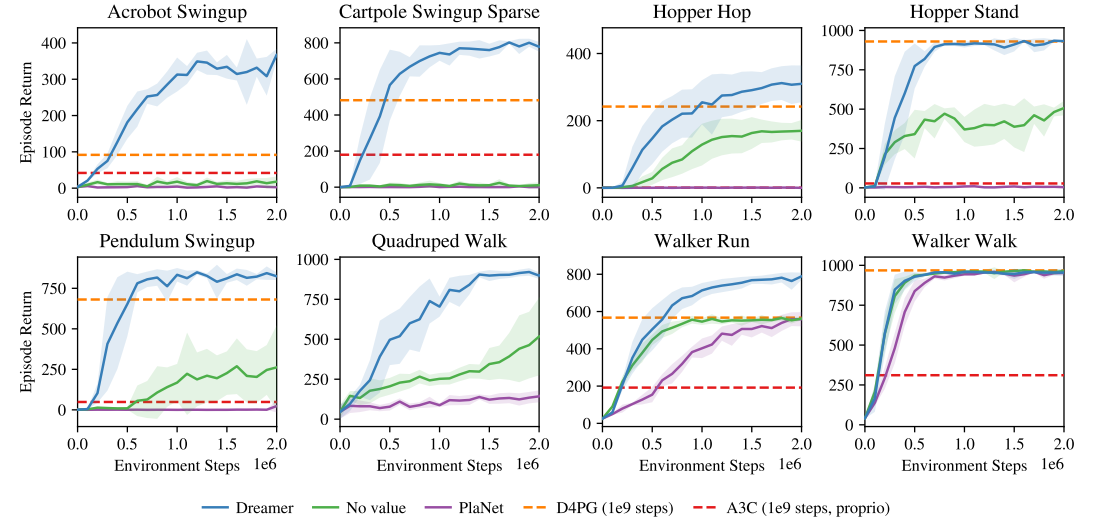


Figure 7: Dreamer succeeds at visual control tasks that require long-horizon credit assignment, such as the acrobot and hopper tasks. Optimizing only imagined rewards within the horizon via an action model or by online planning yields shortsighted behaviors that only succeed in reactive tasks, such as in the walker domain. The performance on all 20 tasks is summarized in Figure 6 and training curves are shown in Appendix D. See Tassa et al. (2018) for performance curves of D4PG and A3C.

5 RELATED WORK

Prior works learn latent dynamics for visual control by derivative-free policy learning or online planning, augment model-free agents with multi-step predictions, or use analytic gradients of Q-values or multi-step rewards, often for low-dimensional tasks. In comparison, Dreamer uses analytic gradients to efficiently learn long-horizon behaviors for visual control purely by latent imagination.

Control with latent dynamics E2C (Watter et al., 2015) and RCE (Banijamali et al., 2017) embed images to predict forward in a compact space to solve simple tasks. World Models (Ha and Schmidhuber, 2018) learn latent dynamics in a two-stage process to evolve linear controllers in imagination. PlaNet (Hafner et al., 2018) learns them jointly and solves visual locomotion tasks by latent online planning. SOLAR (Zhang et al., 2019) solves robotic tasks via guided policy search in latent space. I2A (Weber et al., 2017) hands imagined trajectories to a model-free policy, while Lee et al. (2019) and Gregor et al. (2019) learn belief representations to accelerate model-free agents.

Imagined multi-step returns VPN (Oh et al., 2017), MVE (Feinberg et al., 2018), and STEVE (Buckman et al., 2018) learn dynamics for multi-step Q-learning from a replay buffer. AlphaGo (Silver et al., 2017) combines predictions of actions and state values with planning, assuming access to the true dynamics. Also assuming access to the dynamics, POLO (Lowrey et al., 2018) plans to explore by learning a value ensemble. MuZero (Schrittwieser et al., 2019) learns task-specific reward and value models to solve challenging tasks but requires large amounts of experience. PETS (Chua et al., 2018), VisualMPC (Ebert et al., 2017), and PlaNet (Hafner et al., 2018) plan online using derivative-free optimization. POPLIN (Wang and Ba, 2019) improves over online planning by self-imitation. Piergiovanni et al. (2018) learn robot policies by imagination with a latent dynamics model. Planning with neural network gradients was shown on small problems (Schmidhuber, 1990; Henaff et al., 2018) but has been challenging to scale (Parmas et al., 2019).

Analytic value gradients DPG (Silver et al., 2014), DDPG (Lillicrap et al., 2015), and SAC (Haarnoja et al., 2018) leverage gradients of learned immediate action values to learn a policy by experience replay. SVG (Heess et al., 2015) reduces the variance of model-free on-policy algorithms by analytic value gradients of one-step model predictions. Concurrent work by Byravan et al. (2019) uses latent imagination with deterministic models for navigation and manipulation tasks. ME-TRPO (Kurutach et al., 2018) accelerates an otherwise model-free agent via gradients of predicted rewards for proprioceptive inputs. DistGBP (Henaff et al., 2017; 2019) uses model gradients for online planning in simple tasks.

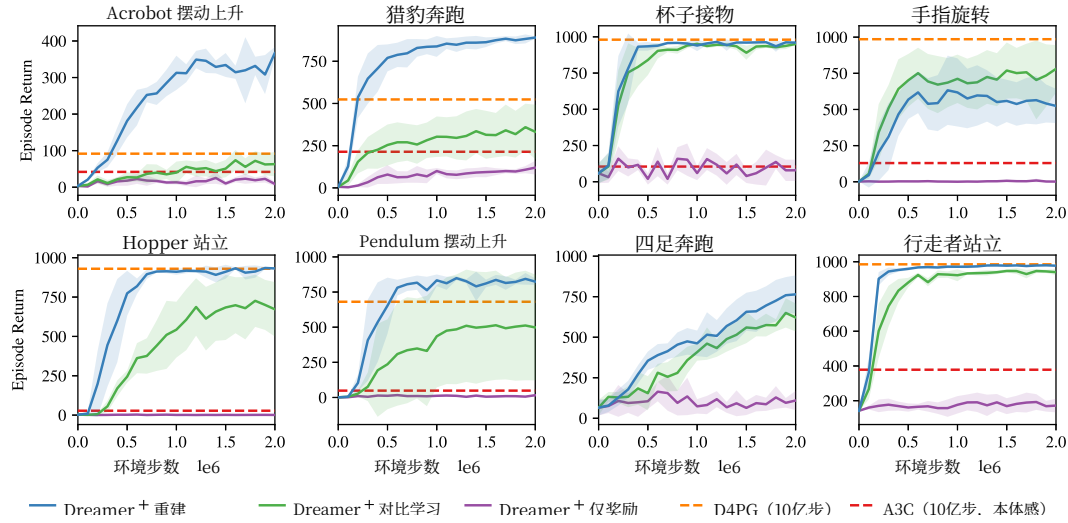


图8：与Dreamer配合使用的表征学习目标对比。对于大多数任务，像素重建表现最佳。对比学习目标可解决约一半的任务，而仅预测奖励在我们的实验中并不足够。结果表明，未来在表征学习方面的发展可能会转化为Dreamer任务性能的提升。所有任务的性能曲线详见附录E。

6 实验

我们在多种控制任务上对 Dreamer 进行了实验评估。我们设计了这些实验，旨在将 Dreamer 与文献中当前最佳方法进行比较，并评估其解决具有长视野、连续动作、离散动作和提前终止的任务的能力。我们还进一步比较了世界模型学习目标的正交选择。所有实验的源代码以及 Dreamer 的视频均可访问<https://danijar.com/dreamer>获取。

控制任务 我们在DeepMind控制套件（Tassa等人, 2018）的20个视觉控制任务上评估Dreamer，如图2所示。这些任务提出了各种挑战，包括稀疏奖励、接触动力学和3D场景。我们选择了Tassa等人（2018）报告从图像输入中获得非零性能的任务。智能体的观测是形状为 $64 \times 64 \times 3$ 的图像，动作维度从1到12，奖励范围从0到1，每个情节持续1000步，并具有随机初始状态。我们在所有任务中使用固定的动作重复为 $R = 2$ 。我们还进一步在部分Atari游戏（Bellemare等人, 2013）和DeepMind Lab关卡（Beattie等人, 2016）上评估Dreamer对离散动作和提前终止的适用性，详见附录C。

实现细节 我们的实现使用TensorFlow概率库（Dillon等, 2017）。每次训练运行时，我们使用单个Nvidia V100 GPU和10个CPU核心。我们的Dreamer实现训练时间约为每 10^6 环境步骤3小时（在控制套件上），相比使用PlaNet进行在线规划的11小时，以及D4PG达到相似性能所需的24小时。我们在所有连续任务中使用相同的超参数，同样在所有离散任务中也使用相同的超参数，详见附录A。除非另有说明，世界模型均通过重建进行学习。

基线方法 在连续控制任务中，最高报告性能由D4PG（Barth-Maron等人, 2018）取得，该算法是DDPG（Lillicrap等人, 2015）的改进版本，采用了分布式收集、分布式Q学习、多步回报和优先级回放。我们纳入了Tassa等人（2018）研究中基于像素输入的D4PG和基于状态输入的A3C（Mnih等人, 2016）的得分。PlaNet（Hafner等人, 2018）学习与Dreamer相同的世界模型，并通过在线规划选择动作，无需动作模型，在数据效率上大幅优于D4PG和A3C。我们使用 $R = 2$ 重新运行PlaNet以实现统一的实验设置。对于雅达利游戏，我们展示了SimPLe（Kaiser等人, 2019）、DQN（Mnih等人, 2015）和Rainbow（Hessel等人, 2018）在Castro等人（2018）报告中给出的最终性能，而对于DeepMind Lab，则参考了IMPALA（Espeholt等人, 2018）的性能作为基准。

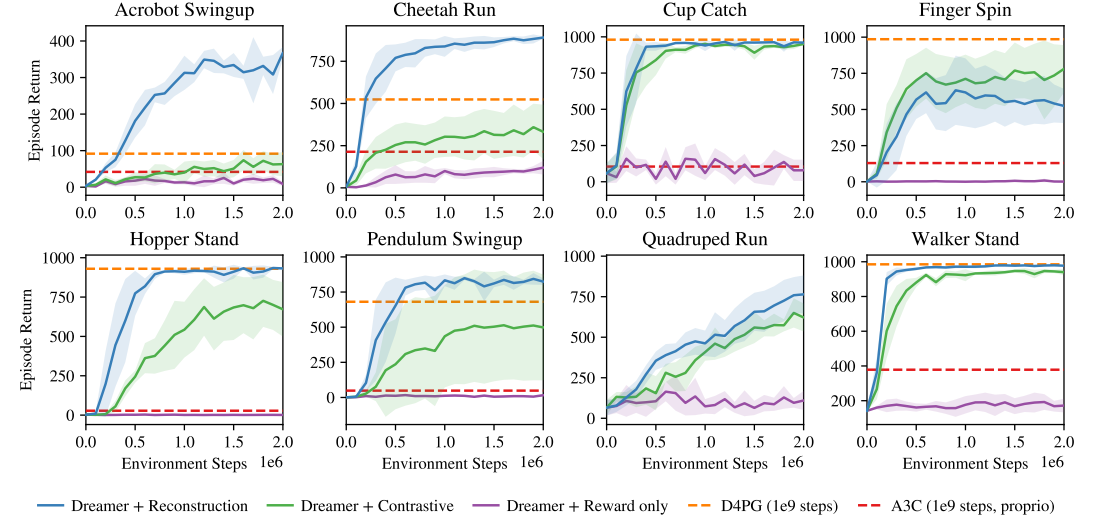


Figure 8: Comparison of representation learning objectives to be used with Dreamer. Pixel reconstruction performs best for the majority of tasks. The contrastive objective solves about half of the tasks, while predicting rewards alone was not sufficient in our experiments. The results suggest that future developments in learning representations are likely to translate into improved task performance for Dreamer. The performance curves for all tasks are included in Appendix E.

6 EXPERIMENTS

We experimentally evaluate Dreamer on a variety of control tasks. We designed the experiments to compare Dreamer to current best methods in the literature, and to evaluate its ability to solve tasks with long horizons, continuous actions, discrete actions, and early termination. We further compare the orthogonal choice of learning objective for the world model. The source code for all our experiments and videos of Dreamer are available at <https://danijar.com/dreamer>.

Control tasks We evaluate Dreamer on 20 visual control tasks of the DeepMind Control Suite (Tassa et al., 2018), illustrated in Figure 2. These tasks pose a variety of challenges, including sparse rewards, contact dynamics, and 3D scenes. We selected the tasks on which Tassa et al. (2018) report non-zero performance from image inputs. Agent observations are images of shape $64 \times 64 \times 3$, actions range from 1 to 12 dimensions, rewards range from 0 to 1, episodes last for 1000 steps and have randomized initial states. We use a fixed action repeat of $R = 2$ across tasks. We further evaluate the applicability of Dreamer to discrete actions and early termination on a subset of Atari games (Bellemare et al., 2013) and DeepMind Lab levels (Beattie et al., 2016) as detailed in Appendix C.

Implementation Our implementation uses TensorFlow Probability (Dillon et al., 2017). We use a single Nvidia V100 GPU and 10 CPU cores for each training run. The training time for our Dreamer implementation is about 3 hours per 10^6 environment steps on the control suite, compared to 11 hours for online planning using PlaNet, and the 24 hours used by D4PG to reach similar performance. We use the same hyper parameters across all continuous tasks, and similarly across all discrete tasks, detailed in Appendix A. The world models are learned via reconstruction unless specified.

Baseline methods The highest reported performance on the continuous tasks is achieved by D4PG (Barth-Maron et al., 2018), an improved variant of DDPG (Lillicrap et al., 2015) that uses distributed collection, distributional Q-learning, multi-step returns, and prioritized replay. We include the scores for D4PG with pixel inputs and A3C (Mnih et al., 2016) with state inputs from Tassa et al. (2018). PlaNet (Hafner et al., 2018) learns the same world model as Dreamer and selects actions via online planning without an action model and drastically improves over D4PG and A3C in data efficiency. We re-run PlaNet with $R = 2$ for a unified experimental setup. For Atari, we show the final performance of SimPLe (Kaiser et al., 2019), DQN (Mnih et al., 2015) and Rainbow (Hessel et al., 2018) reported by Castro et al. (2018), and for DeepMind Lab that of IMPALA (Espeholt et al., 2018) as a guideline.

性能为了评估Dreamer的性能，我们将其与最先进的强化学习智能体进行比较。结果总结于图6。在 5×10^6 环境步数后，Dreamer在各任务上的平均得分为823，超过了强大的无模型D4PG智能体在 10^8 环境步数内取得的平均分786。同时，Dreamer继承了PlaNet的数据效率，证实了学习到的世界模型有助于从少量经验中泛化。Dreamer的实验成功表明，通过世界模型进行潜在想象来学习行为，可以超越基于经验回放的顶级方法。

长视野为了探究其学习长时域行为的能力，我们比较了 Dreamer 与其它从世界模型中推导行为的方法，并设置了不同的想象范围。为此，我们学习了一个动作模型，以在没有价值模型的情况下最大化想象奖励，并将其与使用 PlaNet 进行在线规划的方法进行了对比。图4展示了不同想象范围的最终性能，证实了价值模型使 Dreamer 在视野长度方面更具鲁棒性，即使在短视野下也能表现良好。所有19个任务在想象范围为20时的性能曲线见附录D，其中 Dreamer 在20个任务中的16个上优于其他方法，另有4个任务持平。

表征学习Dreamer 可配合任何可微动力学模型使用，该模型需根据动作和过去观测预测未来奖励。由于表征学习目标与我们的算法正交，我们比较了第4节中描述的三种自然选择：像素重建、对比估计和纯奖励预测。图8显示了不同表征学习方法在任务性能上的显著差异，其中像素重建在大多数任务上优于对比估计。这表明，未来表征学习的改进很可能转化为 Dreamer 更高的任务性能。在我们的实验中，仅靠奖励预测是不够的。进一步的消融研究包含在论文附录中。

7 结论

我们提出了 Dreamer，这是一种完全通过潜在想象来学习长时域行为的智能体。为此，我们提出了一种演员-评论家方法，通过将多步价值的解析梯度反向传播通过已学习的潜在动力学，来优化参数化策略。在多项具有挑战性的图像输入连续控制任务中，Dreamer 在数据效率、计算时间和最终性能方面均优于以往方法。我们进一步证明，Dreamer 也适用于具有离散动作和早期回合终止的任务。未来关于表征学习的研究有望将潜在想象扩展到视觉复杂度更高的环境中。

致谢我们感谢 Simon Kornblith、Benjamin Eysenbach、Ian Fischer、Amy Zhang、Geoffrey Hinton、Shane Gu、Adam Kosiorek、Brandon Amos、Jacob Buckman、Calvin Luo、Rishabh Agarwal 以及匿名审稿人的反馈与讨论。感谢 Yuval Tassa 在控制套件中添加了四足环境。

Performance To evaluate the performance of Dreamer, we compare it to state-of-the-art reinforcement learning agents. The results are summarized in Figure 6. With an average score of 823 across tasks after 5×10^6 environment steps, Dreamer exceeds the performance of the strong model-free D4PG agent that achieves an average of 786 within 10^8 environment steps. At the same time, Dreamer inherits the data-efficiency of PlaNet, confirming that the learned world model can help to generalize from small amounts of experience. The empirical success of Dreamer shows that learning behaviors by latent imagination with world models can outperform top methods based on experience replay.

Long horizons To investigate its ability to learn long-horizon behaviors, we compare Dreamer to alternatives for deriving behaviors from the world model at various horizon lengths. For this, we learn an action model to maximize imagined rewards without a value model and compare to online planning using PlaNet. Figure 4 shows the final performance for different imagination horizons, confirming that the value model makes Dreamer more robust to the horizon and performs well even for short horizons. Performance curves for all 19 tasks with horizon of 20 are shown in Appendix D, where Dreamer outperforms the alternatives on 16 of 20 tasks, with 4 ties.

Representation learning Dreamer can be used with any differentiable dynamics model that predicts future rewards given actions and past observations. Since the representation learning objective is orthogonal to our algorithm, we compare three natural choices described in Section 4: pixel reconstruction, contrastive estimation, and pure reward prediction. Figure 8 shows clear differences in task performance for different representation learning approaches, with pixel reconstruction outperforming contrastive estimation on most tasks. This suggests that future improvements in representation learning are likely to translate to higher task performance with Dreamer. Reward prediction alone was not sufficient in our experiments. Further ablations are included in the appendix of the paper.

7 CONCLUSION

We present Dreamer, an agent that learns long-horizon behaviors purely by latent imagination. For this, we propose an actor critic method that optimizes a parametric policy by propagating analytic gradients of multi-step values back through learned latent dynamics. Dreamer outperforms previous methods in data-efficiency, computation time, and final performance on a variety of challenging continuous control tasks with image inputs. We further show that Dreamer is applicable to tasks with discrete actions and early episode termination. Future research on representation learning can likely scale latent imagination to environments of higher visual complexity.

Acknowledgements We thank Simon Kornblith, Benjamin Eysenbach, Ian Fischer, Amy Zhang, Geoffrey Hinton, Shane Gu, Adam Kosiorek, Brandon Amos, Jacob Buckman, Calvin Luo, and Rishabh Agarwal, and our anonymous reviewers for feedback and discussions. We thank Yuval Tassa for adding the quadruped environment to the control suite.

参考文献

A. A. Alemi, I. Fischer, J. V. Dillon, 和 K. Murphy. 深度变分信息瓶颈. *arXiv预印本 arXiv:1612.00410*, 2016. E. Banijamali, R. Shu, M. Ghavamzadeh, H. Bui, 和 A. Ghodsi. 鲁棒局部线性可控嵌入. *arXiv预印本 arXiv:1710.05373*, 2017. G. Barth-Maron, M. W. Hoffman, D. Budden, W. Dabney, D. Horgan, A. Muldal, N. Heess, 和 T. Lil-licrap. 分布式分布确定性策略梯度. *arXiv预印本 arXiv:1804.08617*, 2018. C. Beattie, J. Z. Leibo, D. Teplyashin, T. Ward, M. Wainwright, H. Küttler, A. Lefrancq, S. Green, V. Valdés, A. Sadik, 等. DeepMind Lab. *arXiv预印本 arXiv:1612.03801*, 2016. M. G. Bellemare, Y. Naddaf, J. Veness, 和 M. Bowling. 街机学习环境: 通用智能体的评估平台. 人工智能研究期刊, 47:253–279, 2013. Y. Bengio, N. Léonard, 和 A. Courville. 通过随机神经元估计或传播梯度用于条件计算. *arXiv预印本 arXiv:1308.3432*, 2013. J. Buckman, D. Hafner, G. Tucker, E. Brevdo, 和 H. Lee. 基于随机集成值扩展的样本高效强化学习. 在神经信息处理系统进展, 页码 8224–8234, 2018. L. Buesing, T. Weber, S. Racaniere, S. Eslami, D. Rezende, D. P. Reichert, F. Viola, F. Besse, K. Gregor, D. Hassabis, 等. 用于强化学习的快速生成模型的学习与查询. *arXiv预印本 arXiv:1802.03006*, 2018. A. Byravan, J. T. Springenberg, A. Abdolmaleki, R. Hafner, M. Neunert, T. Lampe, N. Siegel, N. Heess, 和 M. Riedmiller. 想象值梯度: 基于模型策略优化与可迁移隐动态模型. *arXiv预印本 arXiv:1910.04142*, 2019. P. S. Castro, S. Moitra, C. Gelada, S. Kumar, 和 M. G. Bellemare. Dopamine: 深度强化学习的研究框架. *arXiv预印本 arXiv:1812.06110*, 2018. K. Chua, R. Calandra, R. McAllister, 和 S. Levine. 使用概率动力学模型在少数试验中进行深度强化学习. 在 神经信息处理系统进展, 页码 4754–4765, 2018. D.-A. Clevert, T. Unterthiner, 和 S. Hochreiter. 通过指数线性单元 (ELU) 实现快速准确的深度网络学习. *arXiv预印本 arXiv:1511.07289*, 2015. J. V. Dillon, I. Langmore, D. Tran, E. Brevdo, S. Vasudevan, D. Moore, B. Patton, A. Alemi, M. Hoffman, 和 R. A. Saurous. TensorFlow分布. *arXiv预印本 arXiv:1711.10604*, 2017. A. Doerr, C. Daniel, M. Schiegg, D. Nguyen-Tuong, S. Schaal, M. Toussaint, 和 S. Trimpe. 概率递归状态空间模型. *arXiv预印本 arXiv:1801.10395*, 2018. F. Ebert, C. Finn, A. X. Lee, 和 S. Levine. 基于时间跳跃连接的自监督视觉规划. *arXiv预印本 arXiv:1710.05268*, 2017. S. A. Eslami, D. J. Rezende, F. Besse, F. Viola, A. S. Morcos, M. Garnelo, A. Ruderman, A. A. Rusu, I. Danihelka, K. Gregor, 等. 神经场景表征与渲染. 科学, 360(6394): 1204–1210, 2018. L. Espeholt, H. Soyer, R. Munos, K. Simonyan, V. Mnih, T. Ward, Y. Doron, V. Firoiu, T. Harley, I. Dunning, 等. IMPALA: 基于重要性加权演员-学习者架构的可扩展分布式深度强化学习. *arXiv预印本 arXiv:1802.01561*, 2018. V. Feinberg, A. Wan, I. Stoica, M. I. Jordan, J. E. Gonzalez, 和 S. Levine. 基于模型的价值估计用于高效无模型强化学习. *arXiv预印本 arXiv:1803.00101*, 2018.

REFERENCES

A. A. Alemi, I. Fischer, J. V. Dillon, and K. Murphy. Deep variational information bottleneck. *arXiv preprint arXiv:1612.00410*, 2016.

E. Banijamali, R. Shu, M. Ghavamzadeh, H. Bui, and A. Ghodsi. Robust locally-linear controllable embedding. *arXiv preprint arXiv:1710.05373*, 2017.

G. Barth-Maron, M. W. Hoffman, D. Budden, W. Dabney, D. Horgan, A. Muldal, N. Heess, and T. Lil-licrap. Distributed distributional deterministic policy gradients. *arXiv preprint arXiv:1804.08617*, 2018.

C. Beattie, J. Z. Leibo, D. Teplyashin, T. Ward, M. Wainwright, H. Küttler, A. Lefrancq, S. Green, V. Valdés, A. Sadik, et al. Deepmind lab. *arXiv preprint arXiv:1612.03801*, 2016.

M. G. Bellemare, Y. Naddaf, J. Veness, and M. Bowling. The arcade learning environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47:253–279, 2013.

Y. Bengio, N. Léonard, and A. Courville. Estimating or propagating gradients through stochastic neurons for conditional computation. *arXiv preprint arXiv:1308.3432*, 2013.

J. Buckman, D. Hafner, G. Tucker, E. Brevdo, and H. Lee. Sample-efficient reinforcement learning with stochastic ensemble value expansion. In *Advances in Neural Information Processing Systems*, pages 8224–8234, 2018.

L. Buesing, T. Weber, S. Racaniere, S. Eslami, D. Rezende, D. P. Reichert, F. Viola, F. Besse, K. Gregor, D. Hassabis, et al. Learning and querying fast generative models for reinforcement learning. *arXiv preprint arXiv:1802.03006*, 2018.

A. Byravan, J. T. Springenberg, A. Abdolmaleki, R. Hafner, M. Neunert, T. Lampe, N. Siegel, N. Heess, and M. Riedmiller. Imagined value gradients: Model-based policy optimization with transferable latent dynamics models. *arXiv preprint arXiv:1910.04142*, 2019.

P. S. Castro, S. Moitra, C. Gelada, S. Kumar, and M. G. Bellemare. Dopamine: A research framework for deep reinforcement learning. *arXiv preprint arXiv:1812.06110*, 2018.

K. Chua, R. Calandra, R. McAllister, and S. Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems*, pages 4754–4765, 2018.

D.-A. Clevert, T. Unterthiner, and S. Hochreiter. Fast and accurate deep network learning by exponential linear units (elus). *arXiv preprint arXiv:1511.07289*, 2015.

J. V. Dillon, I. Langmore, D. Tran, E. Brevdo, S. Vasudevan, D. Moore, B. Patton, A. Alemi, M. Hoffman, and R. A. Saurous. Tensorflow distributions. *arXiv preprint arXiv:1711.10604*, 2017.

A. Doerr, C. Daniel, M. Schiegg, D. Nguyen-Tuong, S. Schaal, M. Toussaint, and S. Trimpe. Probabilistic recurrent state-space models. *arXiv preprint arXiv:1801.10395*, 2018.

F. Ebert, C. Finn, A. X. Lee, and S. Levine. Self-supervised visual planning with temporal skip connections. *arXiv preprint arXiv:1710.05268*, 2017.

S. A. Eslami, D. J. Rezende, F. Besse, F. Viola, A. S. Morcos, M. Garnelo, A. Ruderman, A. A. Rusu, I. Danihelka, K. Gregor, et al. Neural scene representation and rendering. *Science*, 360(6394): 1204–1210, 2018.

L. Espeholt, H. Soyer, R. Munos, K. Simonyan, V. Mnih, T. Ward, Y. Doron, V. Firoiu, T. Harley, I. Dunning, et al. Impala: Scalable distributed deep-rl with importance weighted actor-learner architectures. *arXiv preprint arXiv:1802.01561*, 2018.

V. Feinberg, A. Wan, I. Stoica, M. I. Jordan, J. E. Gonzalez, and S. Levine. Model-based value estimation for efficient model-free reinforcement learning. *arXiv preprint arXiv:1803.00101*, 2018.

C. Gelada, S. Kumar, J. Buckman, O. Nachum, and M. G. Bellemare. Deepmdp: Learning continuous latent space models for representation learning. *arXiv预印本 arXiv:1906.02736*, 2019. K. Gregor, D. J. Rezende, F. Besse, Y. Wu, H. Merzic, and A. v. d. Oord. Shaping belief states with generative environment models for rl. *arXiv预印本 arXiv:1906.09237*, 2019. Z. D. Guo, M. G. Azar, B. Piot, B. A. Pires, T. Pohlen, and R. Munos. Neural predictive belief representations. *arXiv预印本 arXiv:1811.06407*, 2018. M. Gutmann and A. Hyvärinen. Noise-contrastive estimation: A new estimation principle for unnormalized statistical models. In 第十三届国际人工智能与统计会议论文集, pages 297–304, 2010. D. Ha and J. Schmidhuber. World models. *arXiv预印本 arXiv:1803.10122*, 2018. T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. *arXiv预印本 arXiv:1801.01290*, 2018. D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee, and J. Davidson. Learning latent dynamics for planning from pixels. *arXiv预印本 arXiv:1811.04551*, 2018. N. Heess, G. Wayne, D. Silver, T. Lillicrap, T. Erez, and Y. Tassa. Learning continuous control policies by stochastic value gradients. In 神经信息处理系统进展, pages 2944–2952, 2015. M. Henaff, W. F. Whitney, and Y. LeCun. Model-based planning in discrete action spaces. 计算机研究库, abs/1705.07177, 2017. M. Henaff, W. F. Whitney, and Y. LeCun. Model-based planning with discrete and continuous actions. *arXiv预印本 arXiv:1705.07177*, 2018. M. Henaff, A. Canziani, and Y. LeCun. Model-predictive policy learning with uncertainty regularization for driving in dense traffic. *arXiv预印本 arXiv:1901.02705*, 2019. M. Hessel, J. Modayil, H. Van Hasselt, T. Schaul, G. Ostrovski, W. Dabney, D. Horgan, B. Piot, M. Azar, and D. Silver. Rainbow: Combining improvements in deep reinforcement learning. In 第三十二届AAAI人工智能会议, 2018. M. Jaderberg, V. Mnih, W. M. Czarnecki, T. Schaul, J. Z. Leibo, D. Silver, and K. Kavukcuoglu. Reinforcement learning with unsupervised auxiliary tasks. *arXiv预印本 arXiv:1611.05397*, 2016. M. I. Jordan, Z. Ghahramani, T. S. Jaakkola, and L. K. Saul. An introduction to variational methods for graphical models. 机器学习, 37(2):183–233, 1999. L. Kaiser, M. Babaeizadeh, P. Milos, B. Osinski, R. H. Campbell, K. Czechowski, D. Erhan, C. Finn, P. Kozakowski, S. Levine, et al. Model-based reinforcement learning for atari. *arXiv预印本 arXiv:1903.00374*, 2019. R. E. Kalman. A new approach to linear filtering and prediction problems. 基础工程杂志, 82(1):35–45, 1960. M. Karl, M. Soelch, J. Bayer, and P. van der Smagt. Deep variational bayes filters: Unsupervised learning of state space models from raw data. *arXiv预印本 arXiv:1605.06432*, 2016. D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. *arXiv预印本 arXiv:1412.6980*, 2014. D. P. Kingma and M. Welling. Auto-encoding variational bayes. *arXiv预印本 arXiv:1312.6114*, 2013. R. G. Krishnan, U. Shalit, and D. Sontag. Deep kalman filters. *arXiv预印本 arXiv:1511.05121*, 2015. T. Kurutach, I. Clavera, Y. Duan, A. Tamar, and P. Abbeel. Model-ensemble trust-region policy optimization. *arXiv预印本 arXiv:1802.10592*, 2018.

C. Gelada, S. Kumar, J. Buckman, O. Nachum, and M. G. Bellemare. Deepmdp: Learning continuous latent space models for representation learning. *arXiv preprint arXiv:1906.02736*, 2019. K. Gregor, D. J. Rezende, F. Besse, Y. Wu, H. Merzic, and A. v. d. Oord. Shaping belief states with generative environment models for rl. *arXiv preprint arXiv:1906.09237*, 2019. Z. D. Guo, M. G. Azar, B. Piot, B. A. Pires, T. Pohlen, and R. Munos. Neural predictive belief representations. *arXiv preprint arXiv:1811.06407*, 2018. M. Gutmann and A. Hyvärinen. Noise-contrastive estimation: A new estimation principle for unnormalized statistical models. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, pages 297–304, 2010. D. Ha and J. Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018. T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. *arXiv preprint arXiv:1801.01290*, 2018. D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee, and J. Davidson. Learning latent dynamics for planning from pixels. *arXiv preprint arXiv:1811.04551*, 2018. N. Heess, G. Wayne, D. Silver, T. Lillicrap, T. Erez, and Y. Tassa. Learning continuous control policies by stochastic value gradients. In *Advances in Neural Information Processing Systems*, pages 2944–2952, 2015. M. Henaff, W. F. Whitney, and Y. LeCun. Model-based planning in discrete action spaces. *CoRR*, abs/1705.07177, 2017. M. Henaff, W. F. Whitney, and Y. LeCun. Model-based planning with discrete and continuous actions. *arXiv preprint arXiv:1705.07177*, 2018. M. Henaff, A. Canziani, and Y. LeCun. Model-predictive policy learning with uncertainty regularization for driving in dense traffic. *arXiv preprint arXiv:1901.02705*, 2019. M. Hessel, J. Modayil, H. Van Hasselt, T. Schaul, G. Ostrovski, W. Dabney, D. Horgan, B. Piot, M. Azar, and D. Silver. Rainbow: Combining improvements in deep reinforcement learning. In *Thirty-Second AAAI Conference on Artificial Intelligence*, 2018. M. Jaderberg, V. Mnih, W. M. Czarnecki, T. Schaul, J. Z. Leibo, D. Silver, and K. Kavukcuoglu. Reinforcement learning with unsupervised auxiliary tasks. *arXiv preprint arXiv:1611.05397*, 2016. M. I. Jordan, Z. Ghahramani, T. S. Jaakkola, and L. K. Saul. An introduction to variational methods for graphical models. *Machine learning*, 37(2):183–233, 1999. L. Kaiser, M. Babaeizadeh, P. Milos, B. Osinski, R. H. Campbell, K. Czechowski, D. Erhan, C. Finn, P. Kozakowski, S. Levine, et al. Model-based reinforcement learning for atari. *arXiv preprint arXiv:1903.00374*, 2019. R. E. Kalman. A new approach to linear filtering and prediction problems. *Journal of basic Engineering*, 82(1):35–45, 1960. M. Karl, M. Soelch, J. Bayer, and P. van der Smagt. Deep variational bayes filters: Unsupervised learning of state space models from raw data. *arXiv preprint arXiv:1605.06432*, 2016. D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014. D. P. Kingma and M. Welling. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013. R. G. Krishnan, U. Shalit, and D. Sontag. Deep kalman filters. *arXiv preprint arXiv:1511.05121*, 2015. T. Kurutach, I. Clavera, Y. Duan, A. Tamar, and P. Abbeel. Model-ensemble trust-region policy optimization. *arXiv preprint arXiv:1802.10592*, 2018.

Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel. Backpropagation applied to handwritten zip code recognition. 神经计算, 1(4):541–551, 1989. A. X. Lee, A. Nagabandi, P. Abbeel, and S. Levine. Stochastic latent actor-critic: Deep reinforcement learning with a latent variable model. *arXiv预印本 arXiv:1907.00953*, 2019. T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra. Continuous control with deep reinforcement learning. *arXiv预印本 arXiv:1509.02971*, 2015. K. Lowrey, A. Rajeswaran, S. Kakade, E. Todorov, and I. Mordatch. Plan online, learn offline: Efficient learning and exploration via model-based control. *arXiv预印本 arXiv:1811.01848*, 2018. M. C. Machado, M. G. Bellemare, E. Talvitie, J. Veness, M. Hausknecht, and M. Bowling. Revisiting the arcade learning environment: Evaluation protocols and open problems for general agents. 人工智能研究期刊, 61:523–562, 2018. D. McAllester and K. Statos. Formal limitations on the measurement of mutual information. *arXiv预印本 arXiv:1811.04251*, 2018. V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Ried- miller, A. K. Fidjeland, G. Ostrovski, et al. Human-level control through deep reinforcement learning. 自然, 518(7540):529, 2015. V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu. Asynchronous methods for deep reinforcement learning. In 国际机器学习大会, pages 1928–1937, 2016. J. Oh, S. Singh, and H. Lee. Value prediction network. In 神经信息处理系统进展, pages 6118–6128, 2017. A. v. d. Oord, Y. Li, and O. Vinyals. Representation learning with contrastive predictive coding. *arXiv预印本 arXiv:1807.03748*, 2018. P. Parmas, C. E. Rasmussen, J. Peters, and K. Doya. Pippis: Flexible model-based policy search robust to the curse of chaos. *arXiv预印本 arXiv:1902.01240*, 2019. A. Piergiovanni, A. Wu, and M. S. Ryoo. Learning real-world robot policies by dreaming. *arXiv预印本 arXiv:1805.07813*, 2018. B. Poole, S. Ozair, A. v. d. Oord, A. A. Alemi, and G. Tucker. On variational bounds of mutual information. *arXiv预印本 arXiv:1905.06922*, 2019. D. J. Rezende, S. Mohamed, and D. Wierstra. Stochastic backpropagation and approximate inference in deep generative models. *arXiv预印本 arXiv:1401.4082*, 2014. J. Schmidhuber. Making the world differentiable: On using self-supervised fully recurrent neural networks for dynamic reinforcement learning and planning in non-stationary environments. 1990. J. Schrittwieser, I. Antonoglou, T. Hubert, K. Simonyan, L. Sifre, S. Schmitt, A. Guez, E. Lockhart, D. Hassabis, T. Graepel, et al. Mastering atari, go, chess and shogi by planning with a learned model. *arXiv预印本 arXiv:1911.08265*, 2019. J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv预印本 arXiv:1707.06347*, 2017. D. Silver, G. Lever, N. Heess, T. Degris, D. Wierstra, and M. Riedmiller. Deterministic policy gradient algorithms. In 第31届国际机器学习大会论文集, 2014. D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton, et al. Mastering the game of go without human knowledge. 自然, 550(7676): 354, 2017.

Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel. Backpropagation applied to handwritten zip code recognition. *Neural computation*, 1(4):541–551, 1989.

A. X. Lee, A. Nagabandi, P. Abbeel, and S. Levine. Stochastic latent actor-critic: Deep reinforcement learning with a latent variable model. *arXiv preprint arXiv:1907.00953*, 2019.

T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra. Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971*, 2015.

K. Lowrey, A. Rajeswaran, S. Kakade, E. Todorov, and I. Mordatch. Plan online, learn offline: Efficient learning and exploration via model-based control. *arXiv preprint arXiv:1811.01848*, 2018.

M. C. Machado, M. G. Bellemare, E. Talvitie, J. Veness, M. Hausknecht, and M. Bowling. Revisiting the arcade learning environment: Evaluation protocols and open problems for general agents. *Journal of Artificial Intelligence Research*, 61:523–562, 2018.

D. McAllester and K. Statos. Formal limitations on the measurement of mutual information. *arXiv preprint arXiv:1811.04251*, 2018.

V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Ried- miller, A. K. Fidjeland, G. Ostrovski, et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529, 2015.

V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu. Asynchronous methods for deep reinforcement learning. In *International Conference on Machine Learning*, pages 1928–1937, 2016.

J. Oh, S. Singh, and H. Lee. Value prediction network. In *Advances in Neural Information Processing Systems*, pages 6118–6128, 2017.

A. v. d. Oord, Y. Li, and O. Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

P. Parmas, C. E. Rasmussen, J. Peters, and K. Doya. Pippis: Flexible model-based policy search robust to the curse of chaos. *arXiv preprint arXiv:1902.01240*, 2019.

A. Piergiovanni, A. Wu, and M. S. Ryoo. Learning real-world robot policies by dreaming. *arXiv preprint arXiv:1805.07813*, 2018.

B. Poole, S. Ozair, A. v. d. Oord, A. A. Alemi, and G. Tucker. On variational bounds of mutual information. *arXiv preprint arXiv:1905.06922*, 2019.

D. J. Rezende, S. Mohamed, and D. Wierstra. Stochastic backpropagation and approximate inference in deep generative models. *arXiv preprint arXiv:1401.4082*, 2014.

J. Schmidhuber. Making the world differentiable: On using self-supervised fully recurrent neural networks for dynamic reinforcement learning and planning in non-stationary environments. 1990.

J. Schrittwieser, I. Antonoglou, T. Hubert, K. Simonyan, L. Sifre, S. Schmitt, A. Guez, E. Lockhart, D. Hassabis, T. Graepel, et al. Mastering atari, go, chess and shogi by planning with a learned model. *arXiv preprint arXiv:1911.08265*, 2019.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

D. Silver, G. Lever, N. Heess, T. Degris, D. Wierstra, and M. Riedmiller. Deterministic policy gradient algorithms. In *Proceedings of the 31st International Conference on Machine Learning*, 2014.

D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton, et al. Mastering the game of go without human knowledge. *Nature*, 550(7676): 354, 2017.

A. Srinivas, A. Jabri, P. Abbeel, S. Levine, 和 C. Finn. 通用规划网络. *arXiv*预印本 *arXiv:1804.00645*, 2018. R. S. Sutton. Dyna: 一种用于学习、规划和反应的集成架构. *ACM SIGART*公报, 2(4):160–163, 1991. R. S. Sutton 和 A. G. Barto. 强化学习导论. 麻省理工学院出版社, 2018. Y. Tassa, Y. Doron, A. Muldal, T. Erez, Y. Li, D. d. L. Casas, D. Budden, A. Abdolmaleki, J. Merel, A. Lefrancq, 等人. DeepMind控制套件. *arXiv*预印本 *arXiv:1801.00690*, 2018. N. Tishby, F. C. Pereira, 和 W. Bialek. 信息瓶颈方法. *arXiv*预印本 *physics/0004057*, 2000. T. Wang 和 J. Ba. 基于策略网络的模型规划探索. *arXiv*预印本 *arXiv:1906.08649*, 2019. T. Wang, X. Bao, I. Clavera, J. Hoang, Y. Wen, E. Langlois, S. Zhang, G. Zhang, P. Abbeel, 和 J. Ba. 基于模型的强化学习基准测试. 计算机研究库, abs/1907.02057, 2019. M. Watter, J. Springenberg, J. Boedecker, 和 M. Riedmiller. 嵌入到控制: 一种用于原始图像控制的局部线性潜在动力学模型. 见神经信息处理系统进展, 页码 2746–2754, 2015. T. Weber, S. Racanière, D. P. Reichert, L. Buesing, A. Guez, D. J. Rezende, A. P. Badia, O. Vinyals, N. Heess, Y. Li, 等人. 想象增强的深度强化学习智能体. *arXiv*预印本 *arXiv:1707.06203*, 2017. R. J. Williams. 连接主义强化学习的简单统计梯度跟踪算法. 机器学习, 8(3-4):229–256, 1992. M. Zhang, S. Vikram, L. Smith, P. Abbeel, M. Johnson, 和 S. Levine. SOLAR: 用于基于模型的强化学习的深度结构化表示. 见国际机器学习大会, 2019.

A. Srinivas, A. Jabri, P. Abbeel, S. Levine, and C. Finn. Universal planning networks. *arXiv preprint arXiv:1804.00645*, 2018.

R. S. Sutton. Dyna, an integrated architecture for learning, planning, and reacting. *ACM SIGART Bulletin*, 2(4):160–163, 1991.

R. S. Sutton and A. G. Barto. *Reinforcement learning: An introduction*. MIT press, 2018.

Y. Tassa, Y. Doron, A. Muldal, T. Erez, Y. Li, D. d. L. Casas, D. Budden, A. Abdolmaleki, J. Merel, A. Lefrancq, et al. Deepmind control suite. *arXiv preprint arXiv:1801.00690*, 2018.

N. Tishby, F. C. Pereira, and W. Bialek. The information bottleneck method. *arXiv preprint physics/0004057*, 2000.

T. Wang and J. Ba. Exploring model-based planning with policy networks. *arXiv preprint arXiv:1906.08649*, 2019.

T. Wang, X. Bao, I. Clavera, J. Hoang, Y. Wen, E. Langlois, S. Zhang, G. Zhang, P. Abbeel, and J. Ba. Benchmarking model-based reinforcement learning. *CoRR*, abs/1907.02057, 2019.

M. Watter, J. Springenberg, J. Boedecker, and M. Riedmiller. Embed to control: A locally linear latent dynamics model for control from raw images. In *Advances in neural information processing systems*, pages 2746–2754, 2015.

T. Weber, S. Racanière, D. P. Reichert, L. Buesing, A. Guez, D. J. Rezende, A. P. Badia, O. Vinyals, N. Heess, Y. Li, et al. Imagination-augmented agents for deep reinforcement learning. *arXiv preprint arXiv:1707.06203*, 2017.

R. J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine learning*, 8(3-4):229–256, 1992.

M. Zhang, S. Vikram, L. Smith, P. Abbeel, M. Johnson, and S. Levine. Solar: deep structured representations for model-based reinforcement learning. In *International Conference on Machine Learning*, 2019.

超参数

模型组件 我们使用了来自Ha和Schmidhuber(2018)的卷积编码器和解码器网络, Hafner等人(2018)提出的RSSM, 并将所有其他函数实现为三个大小为300 的全连接层, 并采用ELU激活函数(Clevert等人, 2015)。潜在空间中的分布是30维的对角高斯分布。动作模型输出一个经缩放因子5缩放的双曲正切均值, 以及一个用于正态分布的Softplus标准差, 然后通过双曲正切进行变换(Haarnoja等人, 2018)。缩放因子允许智能体使动作分布饱和。

学习更新 我们使用Adam (Kingma和Ba, 2014) 以及学习率 $6 \times 10^{-4}, 8 \times 10^{-5}, 8 \times 10^{-5}$ 分别训练世界模型、价值模型和动作模型, 并缩放过100的梯度范数。我们不缩放KL正则化器 ($\beta = 1$), 但将其裁剪至3 自由奈特以下, 如同PlaNet中的做法。想象范围为 $H = 15$, 并且使用相同的轨迹来同时更新动作模型和价值模型。我们使用 V_λ 目标和 $\gamma = 0.99$ 和 $\lambda = 0.95$ 进行计算。我们没有发现学习模型需要进行潜空间过冲、对动作模型施加熵奖励或为价值模型设置目标网络的必要性。

环境交互 数据集使用 $S = 5$ 条由随机动作收集的回合进行初始化。我们在100 训练步数和1 个回合的采集之间迭代, 采集时通过执行模态动作加上Normal(0,0.3) 探索噪声来实现。与Hafner 等人 (2018) 和Lee 等人 (2019) 中为每个环境手动选择动作重复的做法不同, 我们将其固定为所有环境的2。关于不同动作重复值鲁棒性的评估, 请参见图12。

离散控制 在Atari游戏和DeepMind Lab关卡的实验中, 动作模型预测分类分布的logits。在潜在想象过程中, 我们使用直通梯度来处理采样步骤。动作噪声采用 ϵ -贪心策略, 其中 ϵ 在前200,000个梯度步数内从 $0.4 \rightarrow 0.1$ 线性调度。为了应对这些任务更高的复杂度, 我们使用 $H = 10$ 的想象范围, 将KL正则化器缩放 $\beta = 0.1$ 倍, 并使用tanh函数对奖励进行限幅。我们通过一个二分类器从潜在状态预测折扣因子, 该分类器针对0和 γ 的软标签进行训练。

A HYPER PARAMETERS

Model components We use the convolutional encoder and decoder networks from Ha and Schmidhuber (2018), the RSSM of Hafner et al. (2018), and implement all other functions as three dense layers of size 300 with ELU activations (Clevert et al., 2015). Distributions in latent space are 30-dimensional diagonal Gaussians. The action model outputs a tanh mean scaled by a factor of 5 and a softplus standard deviation for the Normal distribution that is then transformed using tanh (Haarnoja et al., 2018). The scaling factor allows the agent to saturate the action distribution.

Learning updates We draw batches of 50 sequences of length 50 to train the world model, value model, and action model models using Adam (Kingma and Ba, 2014) with learning rates $6 \times 10^{-4}, 8 \times 10^{-5}, 8 \times 10^{-5}$, respectively and scale down gradient norms that exceed 100. We do not scale the KL regularizers ($\beta = 1$) but clip them below 3 free nats as in PlaNet. The imagination horizon is $H = 15$ and the same trajectories are used to update both action and value models. We compute the V_λ targets with $\gamma = 0.99$ and $\lambda = 0.95$. We did not find latent overshooting for learning the model, an entropy bonus for the action model, or target networks for the value model necessary.

Environment interaction The dataset is initialized with $S = 5$ episodes collected using random actions. We iterate between 100 training steps and collecting 1 episode by executing the predicted mode action with Normal(0, 0.3) exploration noise. Instead of manually selecting the action repeat for each environment as in Hafner et al. (2018) and Lee et al. (2019), we fix it to 2 for all environments. See Figure 12 for an assessment of the robustness to different action repeat values.

Discrete control For experiments on Atari games and DeepMind Lab levels, the action model predicts the logits of a categorical distribution. We use straight-through gradients for the sampling step during latent imagination. The action noise is epsilon greedy where ϵ is linearly scheduled from $0.4 \rightarrow 0.1$ over the first 200,000 gradient steps. To account for the higher complexity of these tasks, we use an imagination horizon of $H = 10$, scale the KL regularizers by $\beta = 0.1$, and bound rewards using tanh. We predict the discount factor from the latent state with a binary classifier that is trained towards the soft labels of 0 and γ .

B 推导

我们为潜在动力学模型定义了信息瓶颈目标 (Tishby等人,2000)

$$\max \mathbf{I}(s_{1:T}; (o_{1:T}, r_{1:T}) \mid a_{1:T}) - \beta \mathbf{I}(s_{1:T}, i_{1:T} \mid a_{1:T}), \quad (13)$$

其中 β 是标量, i_t 是决定观测值的数据集索引 $p(o_t \mid i_t) \doteq \delta(o_t - \bar{o}_t)$, 如 Alemi 等人 (2016) 所述。最大化该目标函数将得到能预测观测序列和奖励的模型状态, 同时限制每个时间步提取的信息量。这鼓励模型尽可能依赖先前时间步提取的信息来重建每张图像, 仅在必要时从当前图像中获取额外信息。因此, 信息正则化器促使模型学习长期依赖关系。

对于生成目标, 我们利用 KL 散度的非负性对第一项进行下界约束, 并忽略不依赖于表征模型的边际数据概率,

$$\begin{aligned} & \mathbf{I}(s_{1:T}; (o_{1:T}, r_{1:T}) \mid a_{1:T}) \\ = & \mathbb{E}_{p(o_{1:T}, r_{1:T}, s_{1:T}, a_{1:T})} \left(\sum_t \ln p(o_{1:T}, r_{1:T} \mid s_{1:T}, a_{1:T}) - \underbrace{\ln p(o_{1:T}, r_{1:T} \mid a_{1:T})}_{\text{const}} \right) \\ \stackrel{\pm}{=} & \mathbb{E} \left(\sum_t \ln p(o_{1:T}, r_{1:T} \mid s_{1:T}, a_{1:T}) \right) \\ \geq & \mathbb{E} \left(\sum_t \ln p(o_{1:T}, r_{1:T} \mid s_{1:T}, a_{1:T}) \right) - \text{KL} \left(p(o_{1:T}, r_{1:T} \mid s_{1:T}, a_{1:T}) \parallel \prod_t q(o_t \mid s_t) q(r_t \mid s_t) \right) \\ = & \mathbb{E} \left(\sum_t \ln q(o_t \mid s_t) + \ln q(r_t \mid s_t) \right). \end{aligned} \quad (14)$$

For 对比目标, 我们减去数据在该条件下的常数边际概率, 应用贝叶斯规则, 并使用 InfoNCE 小批量边界 (Poole 等人, 2019)

$$\begin{aligned} & \mathbb{E}(\ln q(o_t \mid s_t) + \ln q(r_t \mid s_t)) \\ \stackrel{\pm}{=} & \mathbb{E}(\ln q(o_t \mid s_t) - \ln q(o_t) + \ln q(r_t \mid s_t)) \\ = & \mathbb{E}(\ln q(s_t \mid o_t) - \ln q(s_t) + \ln q(r_t \mid s_t)) \\ \geq & \mathbb{E} \left(\ln q(s_t \mid o_t) - \ln \sum_{o'} q(s_t \mid o') + \ln q(r_t \mid s_t) \right). \end{aligned} \quad (15)$$

对于第二项, 我们利用 KL 散度的非负性得到一个上界,

$$\begin{aligned} & \mathbf{I}(s_{1:T}; i_{1:T} \mid a_{1:T}) \\ = & \mathbb{E}_{p(o_{1:T}, r_{1:T}, s_{1:T}, a_{1:T}, i_{1:T})} \left(\sum_t \ln p(s_t \mid s_{t-1}, a_{t-1}, i_t) - \ln p(s_t \mid s_{t-1}, a_{t-1}) \right) \\ = & \mathbb{E} \left(\sum_t \ln p(s_t \mid s_{t-1}, a_{t-1}, o_t) - \ln p(s_t \mid s_{t-1}, a_{t-1}) \right) \\ \leq & \mathbb{E} \left(\sum_t \ln p(s_t \mid s_{t-1}, a_{t-1}, o_t) - \ln q(s_t \mid s_{t-1}, a_{t-1}) \right) \\ = & \mathbb{E} \left(\sum_t \text{KL} (p(s_t \mid s_{t-1}, a_{t-1}, o_t) \parallel q(s_t \mid s_{t-1}, a_{t-1})) \right). \end{aligned} \quad (16)$$

这为目标函数提供了一个下界。

B DERIVATIONS

We define the information bottleneck objective (Tishby et al., 2000) for latent dynamics models,

$$\max \mathbf{I}(s_{1:T}; (o_{1:T}, r_{1:T}) \mid a_{1:T}) - \beta \mathbf{I}(s_{1:T}, i_{1:T} \mid a_{1:T}), \quad (13)$$

where β is scalar and i_t are dataset indices that determine the observations $p(o_t \mid i_t) \doteq \delta(o_t - \bar{o}_t)$ as in Alemi et al. (2016).

Maximizing the objective leads to model states that can predict the sequence of observations and rewards while limiting the amount of information extracted at each time step. This encourages the model to reconstruct each image by relying on information extracted at preceeding time steps to the extent possible, and only accessing additional information from the current image when necessary. As a result, the information regularizer encourages the model to learn long-term dependencies.

For the generative objective, we lower bound the first term using the non-negativity of the KL divergence and drop the marginal data probability as it does not depend on the representation model,

$$\begin{aligned} & \mathbf{I}(s_{1:T}; (o_{1:T}, r_{1:T}) \mid a_{1:T}) \\ = & \mathbb{E}_{p(o_{1:T}, r_{1:T}, s_{1:T}, a_{1:T})} \left(\sum_t \ln p(o_{1:T}, r_{1:T} \mid s_{1:T}, a_{1:T}) - \underbrace{\ln p(o_{1:T}, r_{1:T} \mid a_{1:T})}_{\text{const}} \right) \\ \stackrel{\pm}{=} & \mathbb{E} \left(\sum_t \ln p(o_{1:T}, r_{1:T} \mid s_{1:T}, a_{1:T}) \right) \\ \geq & \mathbb{E} \left(\sum_t \ln p(o_{1:T}, r_{1:T} \mid s_{1:T}, a_{1:T}) \right) - \text{KL} \left(p(o_{1:T}, r_{1:T} \mid s_{1:T}, a_{1:T}) \parallel \prod_t q(o_t \mid s_t) q(r_t \mid s_t) \right) \\ = & \mathbb{E} \left(\sum_t \ln q(o_t \mid s_t) + \ln q(r_t \mid s_t) \right). \end{aligned} \quad (14)$$

For the contrastive objective, we subtract the constant marginal probability of the data under the variational encoder, apply Bayes rule, and use the InfoNCE mini-batch bound (Poole et al., 2019),

$$\begin{aligned} & \mathbb{E}(\ln q(o_t \mid s_t) + \ln q(r_t \mid s_t)) \\ \stackrel{\pm}{=} & \mathbb{E}(\ln q(o_t \mid s_t) - \ln q(o_t) + \ln q(r_t \mid s_t)) \\ = & \mathbb{E}(\ln q(s_t \mid o_t) - \ln q(s_t) + \ln q(r_t \mid s_t)) \\ \geq & \mathbb{E} \left(\ln q(s_t \mid o_t) - \ln \sum_{o'} q(s_t \mid o') + \ln q(r_t \mid s_t) \right). \end{aligned} \quad (15)$$

For the second term, we use the non-negativity of the KL divergence to obtain an upper bound,

$$\begin{aligned} & \mathbf{I}(s_{1:T}; i_{1:T} \mid a_{1:T}) \\ = & \mathbb{E}_{p(o_{1:T}, r_{1:T}, s_{1:T}, a_{1:T}, i_{1:T})} \left(\sum_t \ln p(s_t \mid s_{t-1}, a_{t-1}, i_t) - \ln p(s_t \mid s_{t-1}, a_{t-1}) \right) \\ = & \mathbb{E} \left(\sum_t \ln p(s_t \mid s_{t-1}, a_{t-1}, o_t) - \ln p(s_t \mid s_{t-1}, a_{t-1}) \right) \\ \leq & \mathbb{E} \left(\sum_t \ln p(s_t \mid s_{t-1}, a_{t-1}, o_t) - \ln q(s_t \mid s_{t-1}, a_{t-1}) \right) \\ = & \mathbb{E} \left(\sum_t \text{KL} (p(s_t \mid s_{t-1}, a_{t-1}, o_t) \parallel q(s_t \mid s_{t-1}, a_{t-1})) \right). \end{aligned} \quad (16)$$

This lower bounds the objective.

C 离散控制

我们在Atari套件 (Bellemare等人,2013) 和DeepMind Lab (Beattie等人, 2016) 的离散动作任务子集上评估Dreamer。虽然纯粹通过世界模型学习的智能体在这些领域尚未具备竞争力 (Kaiser等人, 2019), 但这些任务提供了具有视觉复杂性、稀疏奖励和提前终止的多样化测试平台。智能体观察 $64 \times 64 \times 3$ 图像, 并选择3 到18 个动作之一。对于雅达利, 我们遵循 Machado等人 (2018) 的评估协议, 使用粘性动作。请参考图9了解这些实验。

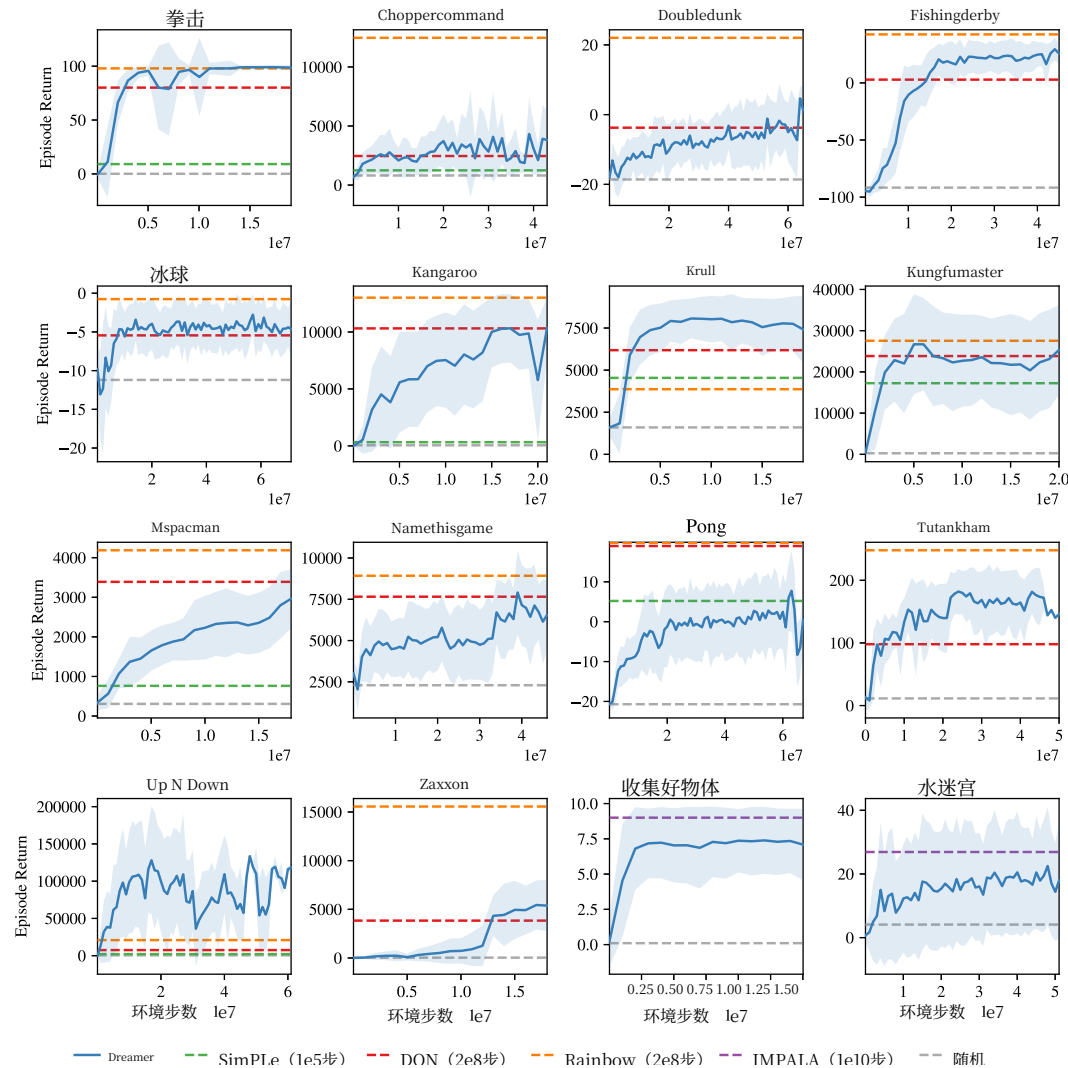


图9: Dreamer在离散动作和提前终止环境中的表现。Dreamer在此类Atari游戏子集及DMLab物体收集关卡中学会了成功的行为。我们将这些环境的表征学习作为未来工作的方向, 以期通过Dreamer在所有Atari游戏和DMLab关卡中实现具有竞争力的表现。

C DISCRETE CONTROL

We evaluate Dreamer on a subset of tasks with discrete actions from the Atari suite (Bellemare et al., 2013) and DeepMind Lab (Beattie et al., 2016). While agents that purely learn through world models are not yet competitive in these domains (Kaiser et al., 2019), the tasks offer a diverse test bed with visual complexity, sparse rewards, and early termination. Agents observe $64 \times 64 \times 3$ images and select one of between 3 and 18 actions. For Atari, we follow the evaluation protocol of Machado et al. (2018) with sticky actions. Refer to Figure 9 for these experiments.

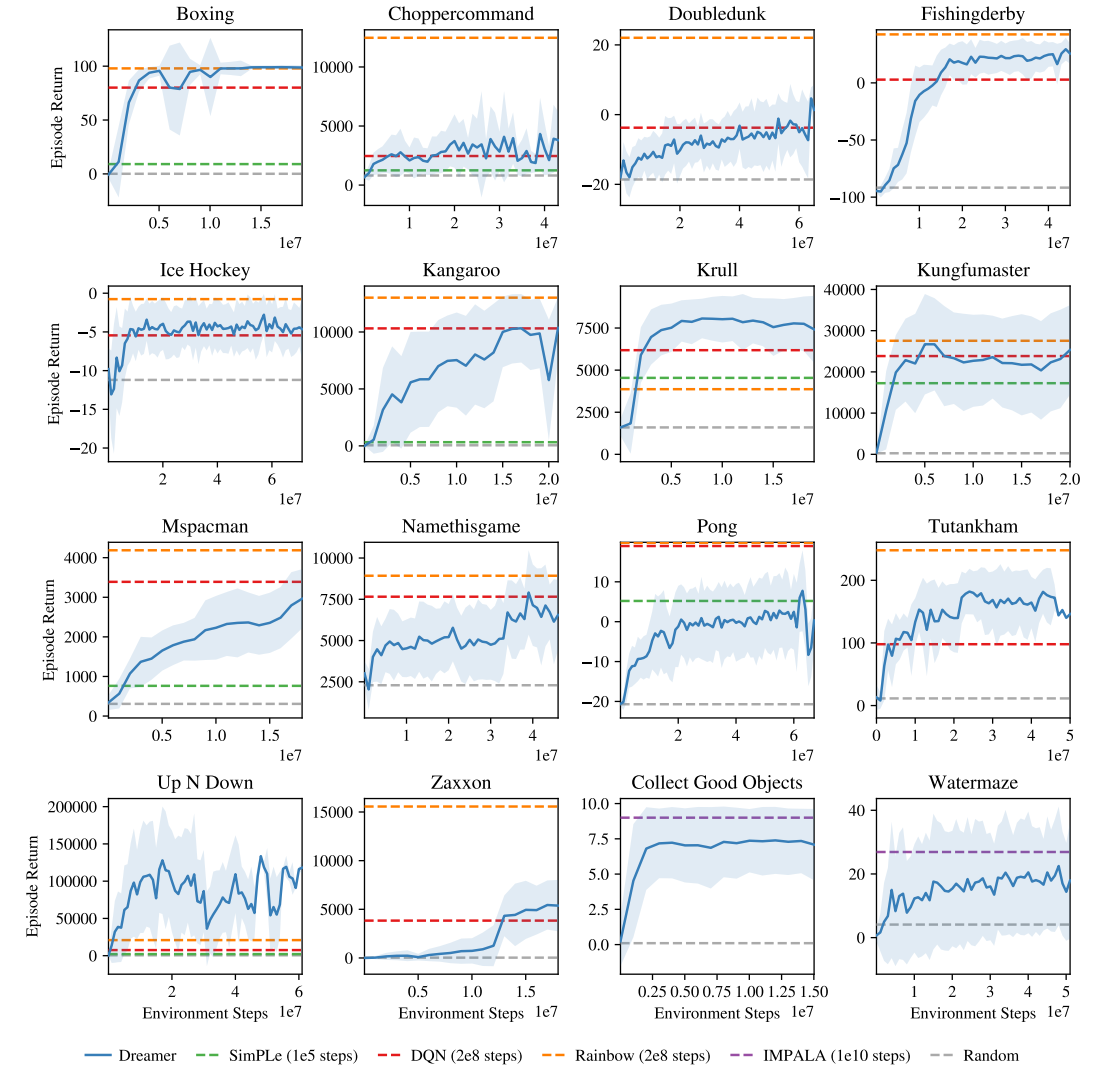


Figure 9: Performance of Dreamer in environments with discrete actions and early termination. Dreamer learns successful behaviors on this subset of Atari games and the object collection level of DMLab. We highlight representation learning for these environments as a direction of future work that could enable competitive performance across all Atari games and DMLab levels using Dreamer.

D 行为学习

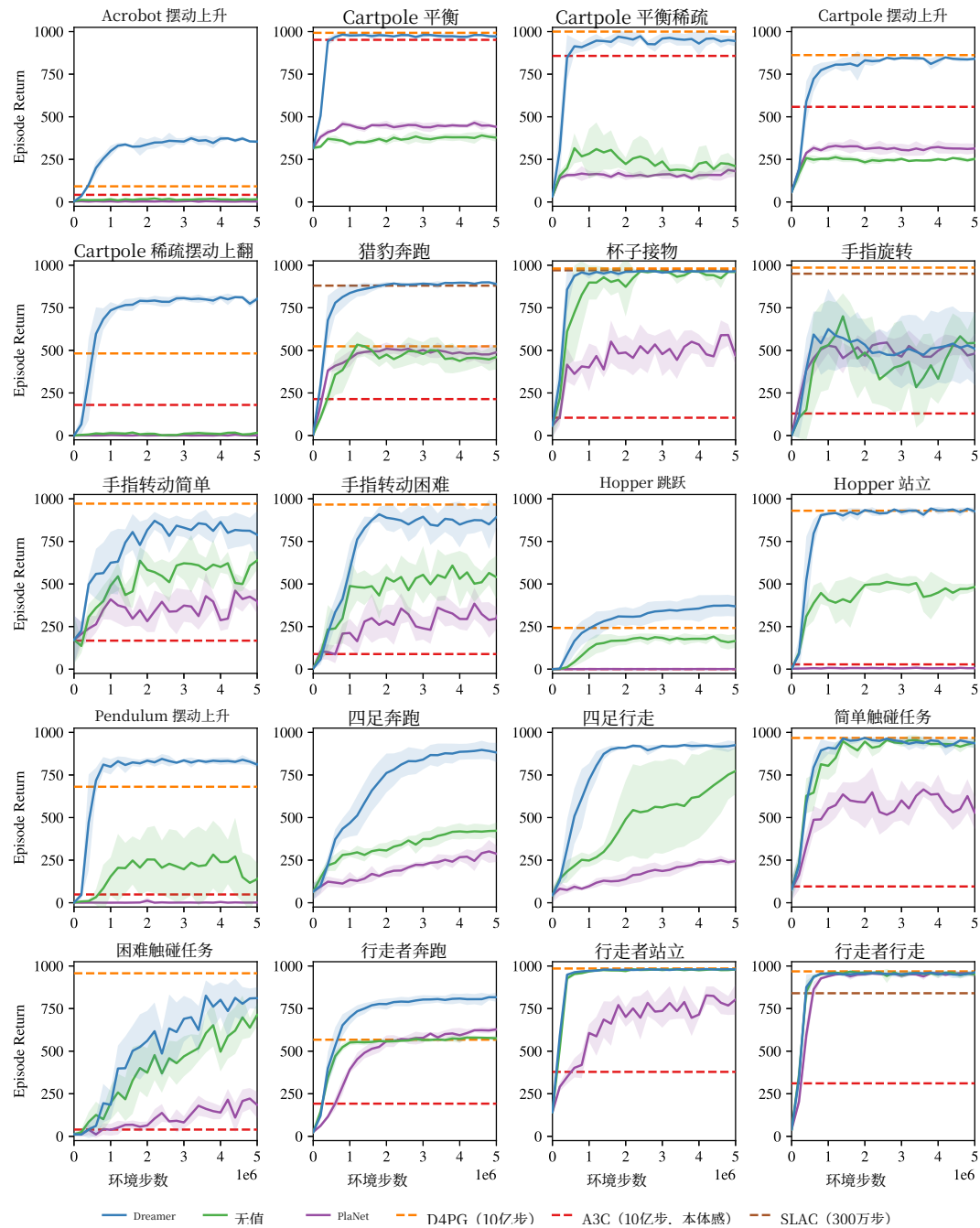


图10: 在 DeepMind 控制套件中基于像素输入的连续控制任务上, 对动作选择方案的比较。曲线显示各环境步数的平均得分, 阴影区域表示 5 个随机种子的标准差。我们将 Dreamer (在想象中同时学习动作与价值) 与仅在想象中学习动作的方案, 以及 PlaNet (通过在线规划选择动作, 而非学习策略) 进行比较。基线包括顶级无模型算法 D4PG、广为人知的 A3C 智能体, 以及混合 SLAC 智能体。

D BEHAVIOR LEARNING

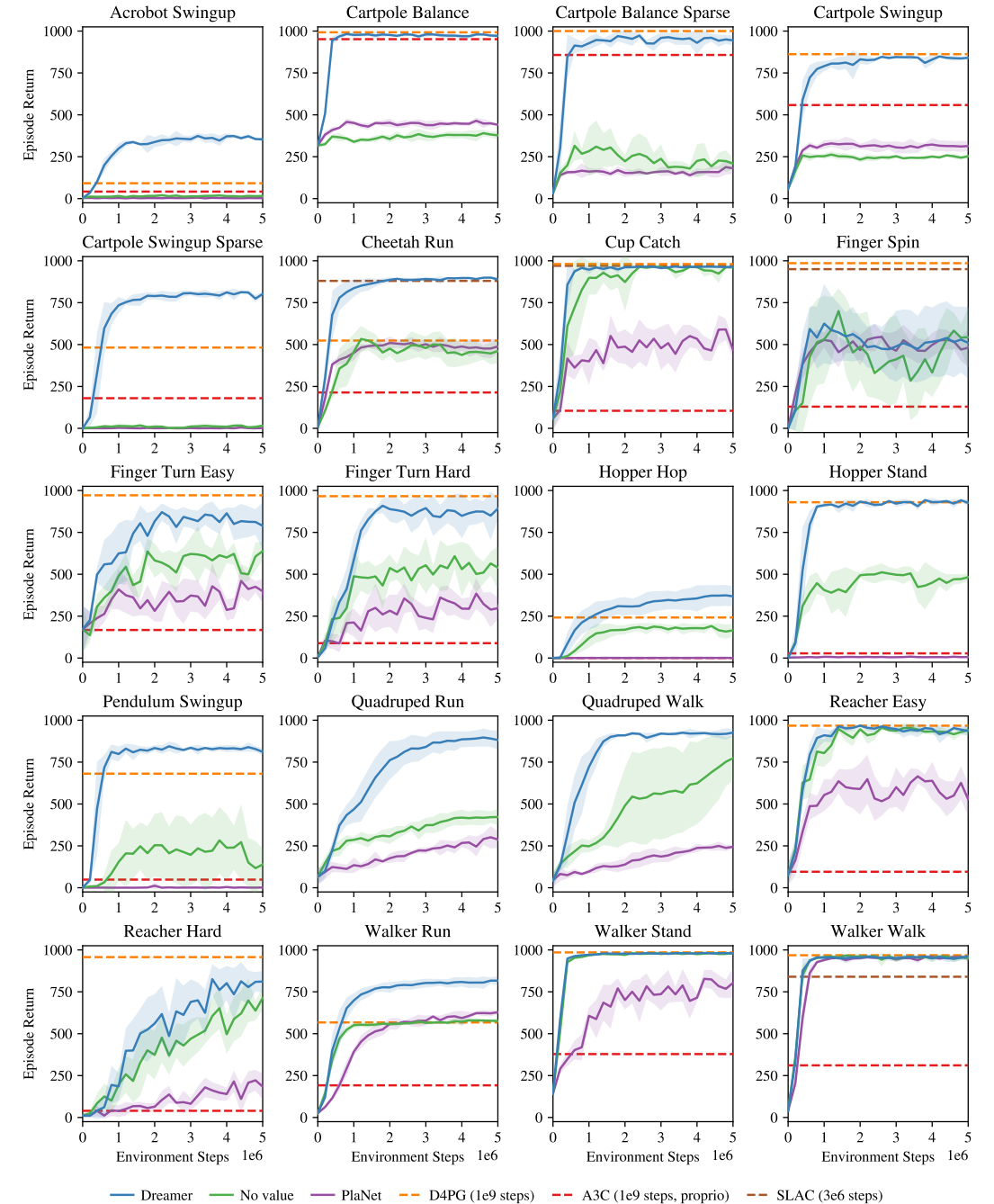


Figure 10: Comparison of action selection schemes on the continuous control tasks of the DeepMind Control Suite from pixel inputs. The lines show mean scores over environment steps and the shaded areas show the standard deviation across 5 seeds. We compare Dreamer that learns both actions and values in imagination, to only learning actions in imagination, and Planet that selects actions by online planning instead of learning a policy. The baselines include the top model-free algorithm D4PG, the well-known A3C agent, and the hybrid SLAC agent.

E 表征学习

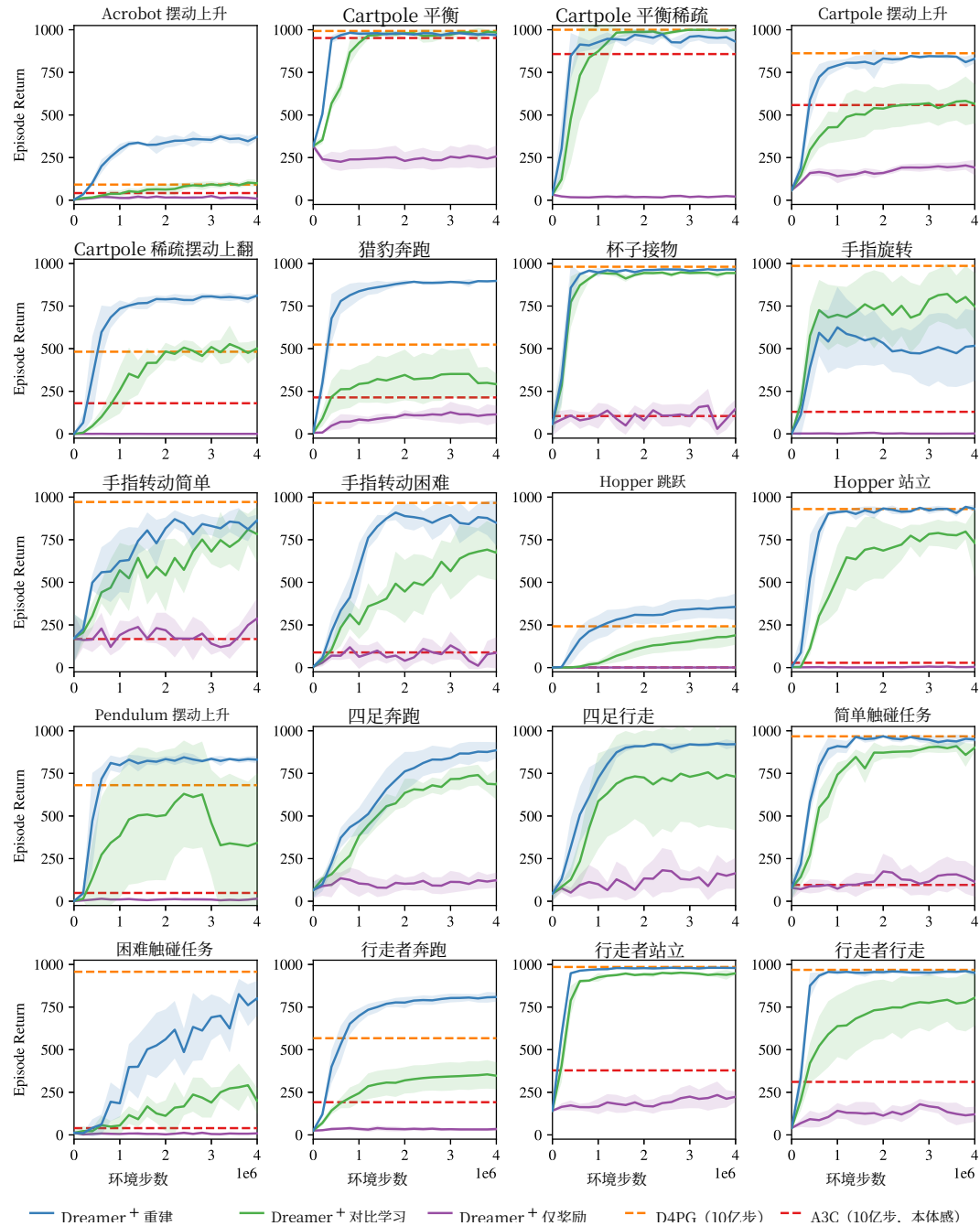


图11: Dreamer的表征学习方法对比。线条表示平均得分, 阴影区域表示5个随机种子下的标准差。我们比较了同时生成图像和奖励、生成奖励并使用对比损失学习图像信息、以及仅预测奖励这三种方法。在大多数任务中, 图像重建提供了最佳的学习信号, 其次是对比目标。在我们的实验中, 仅从奖励中学习是不够的, 可能需要更多的经验积累。

E REPRESENTATION LEARNING

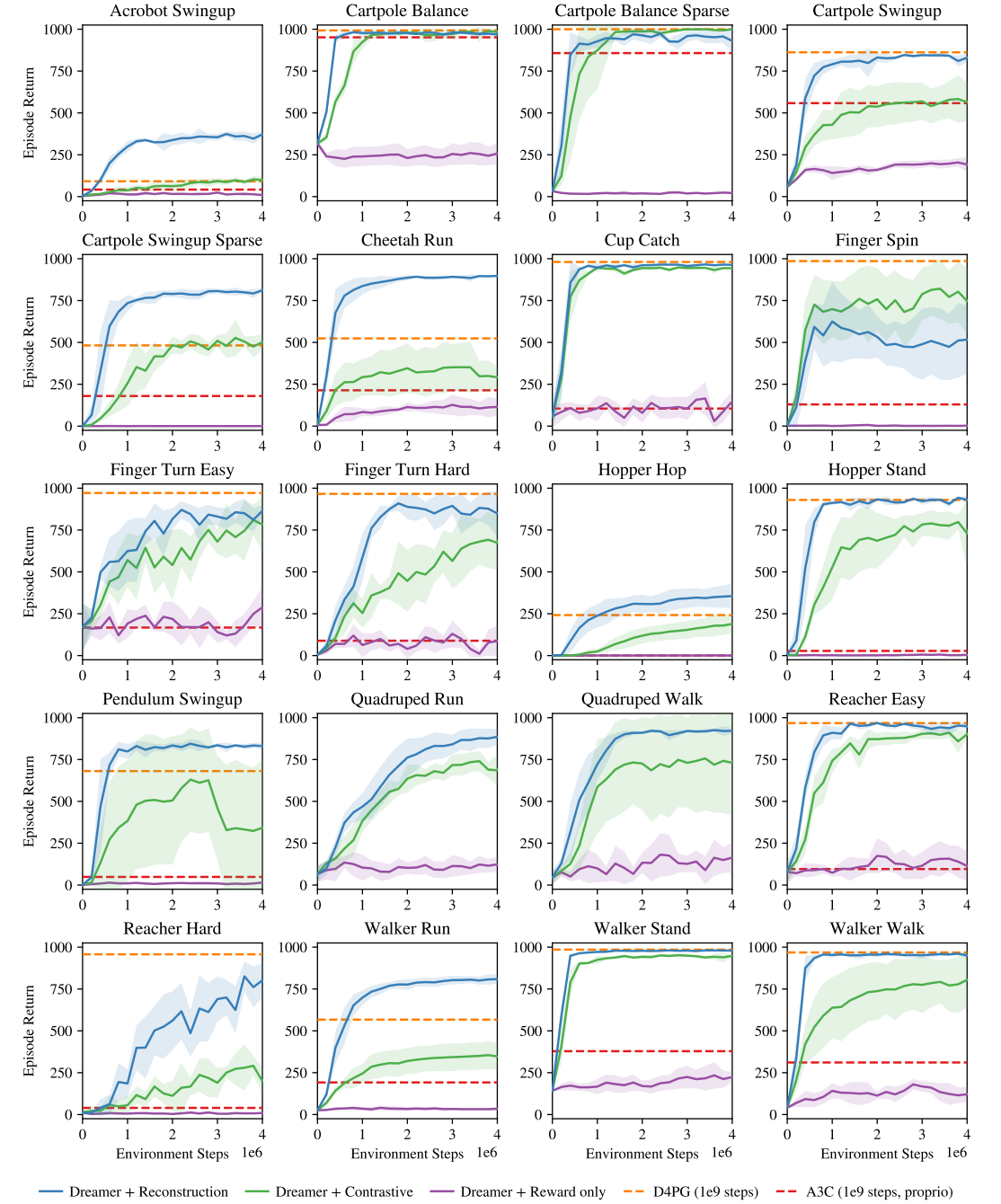


Figure 11: Comparison of representation learning methods for Dreamer. The lines show mean scores and the shaded areas show the standard deviation across 5 seeds. We compare generating both images and rewards, generating rewards and using a contrastive loss to learn about the images, and only predicting rewards. Image reconstruction provides the best learning signal across most of the tasks, followed by the contrastive objective. Learning purely from rewards was not sufficient in our experiments and might require larger amounts of experience.

F 动作重复

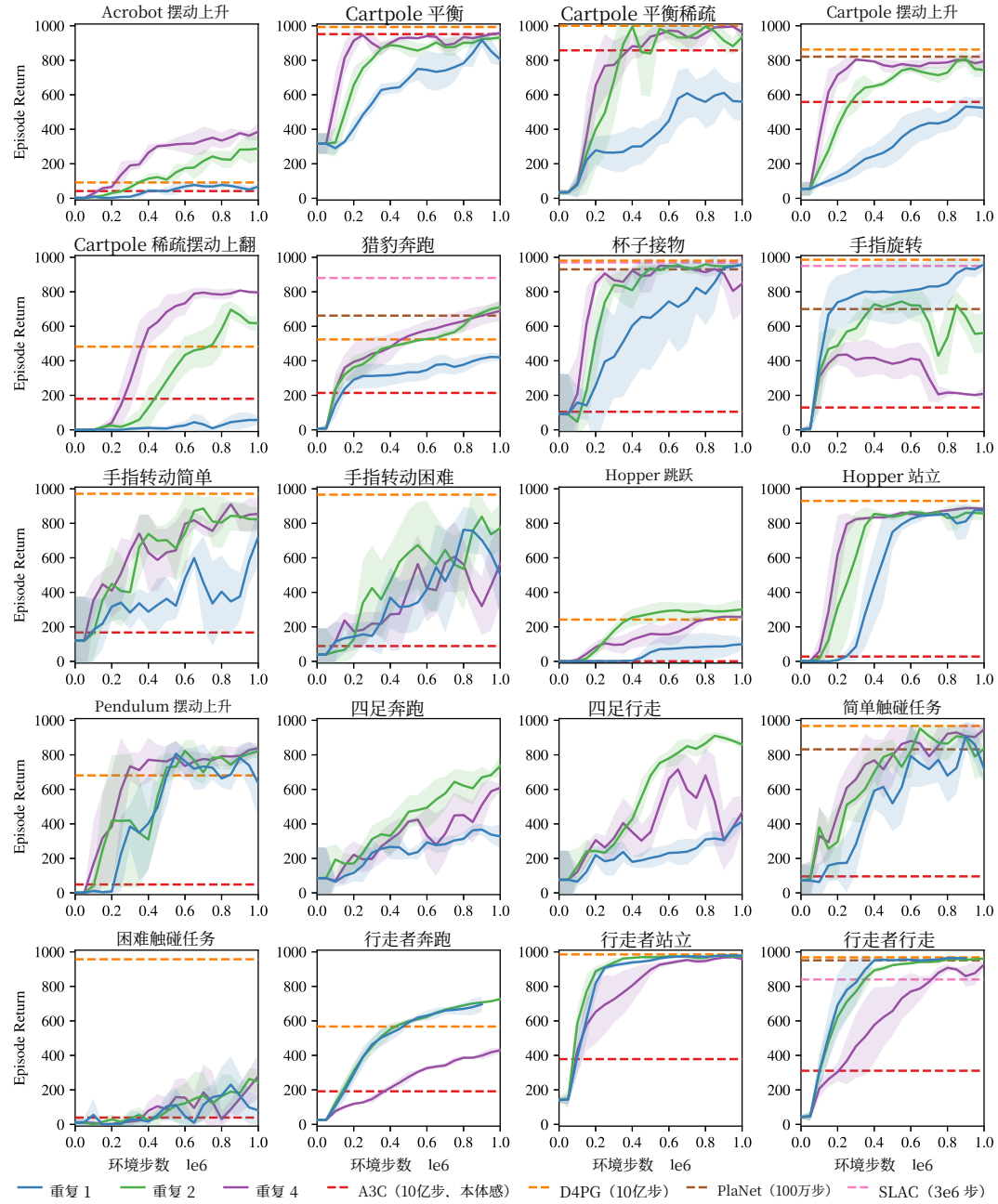


图 12: Dreamer 对不同控制频率的鲁棒性。强化学习方法可能对该超参数敏感, 而在环境的控制频率下学习动力学模型时, 这种敏感性可能会被放大。在此实验中, 我们使用不同的动作重复次数训练 Dreamer。色带区域展示了 2 个随机种子下的一个标准差。我们在此实验中使用了之前的超参数设置。我们发现值 $R = 2$ 在不同任务中表现最佳。

F ACTION REPEAT

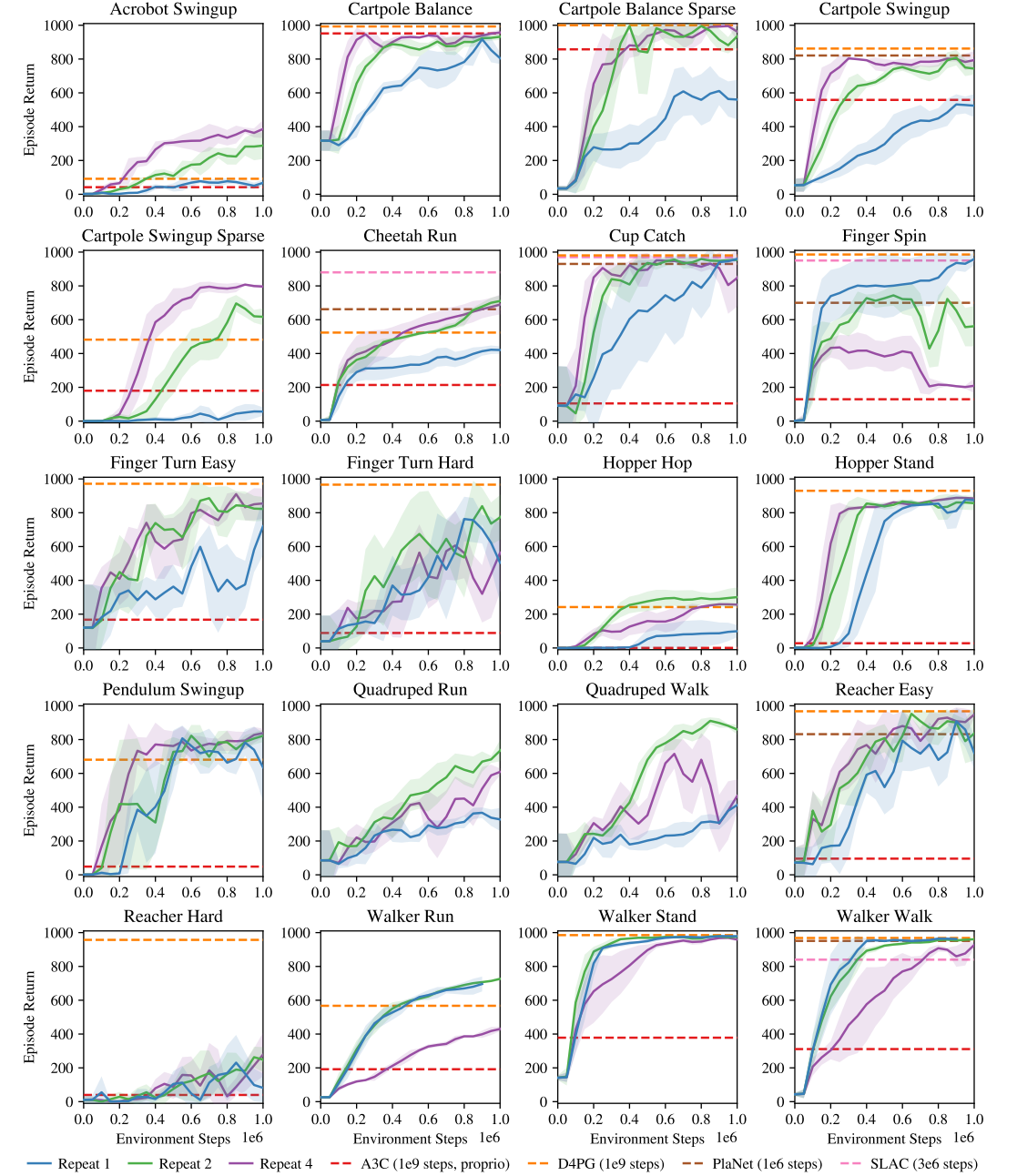


Figure 12: Robustness of Dreamer to different control frequencies. Reinforcement learning methods can be sensitive to this hyper parameter, which could be amplified when learning dynamics models at the control frequency of the environment. For this experiment, we train Dreamer with different amounts of action repeat. The areas show one standard deviation across 2 seeds. We used a previous hyper parameter setting for this experiment. We find that a value of $R = 2$ works best across tasks.

G 连续控制评分

	A3C	D4PG	PlaNet ¹	Dreamer
输入模态	本体感知	像素	像素	像素
环境步数	10^8	10^8	5×10^6	5×10^6
Acrobot 摆动上升	41.90	91.70	3.21	365.26
Cartpole 平衡	951.60	992.80	452.56	979.56
Cartpole 平衡稀疏	857.40	1000.00	164.74	941.84
Cartpole 摆动上升	558.40	862.00	312.56	833.66
Cartpole 稀疏摆动上翻	179.80	482.00	0.64	812.22
猎豹奔跑	213.90	523.80	496.12	894.56
杯子接物	104.70	980.50	455.98	962.48
手指旋转	129.40	985.70	495.25	498.88
手指转动简单	167.30	971.40	451.22	825.86
手指转动困难	88.70	966.00	312.55	891.38
Hopper 跳跃	0.50	242.00	0.37	368.97
Hopper 站立	27.90	929.90	5.96	923.72
Pendulum 摆动上升	48.60	680.90	3.27	833.00
四足奔跑	—	—	280.45	888.39
四足行走	—	—	238.90	931.61
简单触碰任务	95.60	967.40	468.50	935.08
困难触碰任务	39.70	957.10	187.02	817.05
行走者奔跑	191.80	567.20	626.25	824.67
行走者站立	378.40	985.20	759.19	977.99
行走者行走	311.00	968.30	944.70	961.67
平均值	243.70	786.32	332.97	823.39

¹我们使用固定的动作重复率重新运行 PlaNet $R = 2$ ，以避免针对 20 个任务分别调整该值。因此，得分与 [Hafner 等人 \(2018\)](#) 的结果存在差异。

G CONTINUOUS CONTROL SCORES

	A3C	D4PG	PlaNet ¹	Dreamer
Input modality	proprio	pixels	pixels	pixels
Environment steps	10^8	10^8	5×10^6	5×10^6
Acrobot Swingup	41.90	91.70	3.21	365.26
Cartpole Balance	951.60	992.80	452.56	979.56
Cartpole Balance Sparse	857.40	1000.00	164.74	941.84
Cartpole Swingup	558.40	862.00	312.56	833.66
Cartpole Swingup Sparse	179.80	482.00	0.64	812.22
Cheetah Run	213.90	523.80	496.12	894.56
Cup Catch	104.70	980.50	455.98	962.48
Finger Spin	129.40	985.70	495.25	498.88
Finger Turn Easy	167.30	971.40	451.22	825.86
Finger Turn Hard	88.70	966.00	312.55	891.38
Hopper Hop	0.50	242.00	0.37	368.97
Hopper Stand	27.90	929.90	5.96	923.72
Pendulum Swingup	48.60	680.90	3.27	833.00
Quadruped Run	—	—	280.45	888.39
Quadruped Walk	—	—	238.90	931.61
Reacher Easy	95.60	967.40	468.50	935.08
Reacher Hard	39.70	957.10	187.02	817.05
Walker Run	191.80	567.20	626.25	824.67
Walker Stand	378.40	985.20	759.19	977.99
Walker Walk	311.00	968.30	944.70	961.67
Average	243.70	786.32	332.97	823.39

¹We re-run PlaNet with fixed action repeat of $R = 2$ to not tune the this value for each of the 20 tasks. As a result, the scores differ from [Hafner et al. \(2018\)](#).